

Introduction to High-Performance Machine Learning

Lecture 11

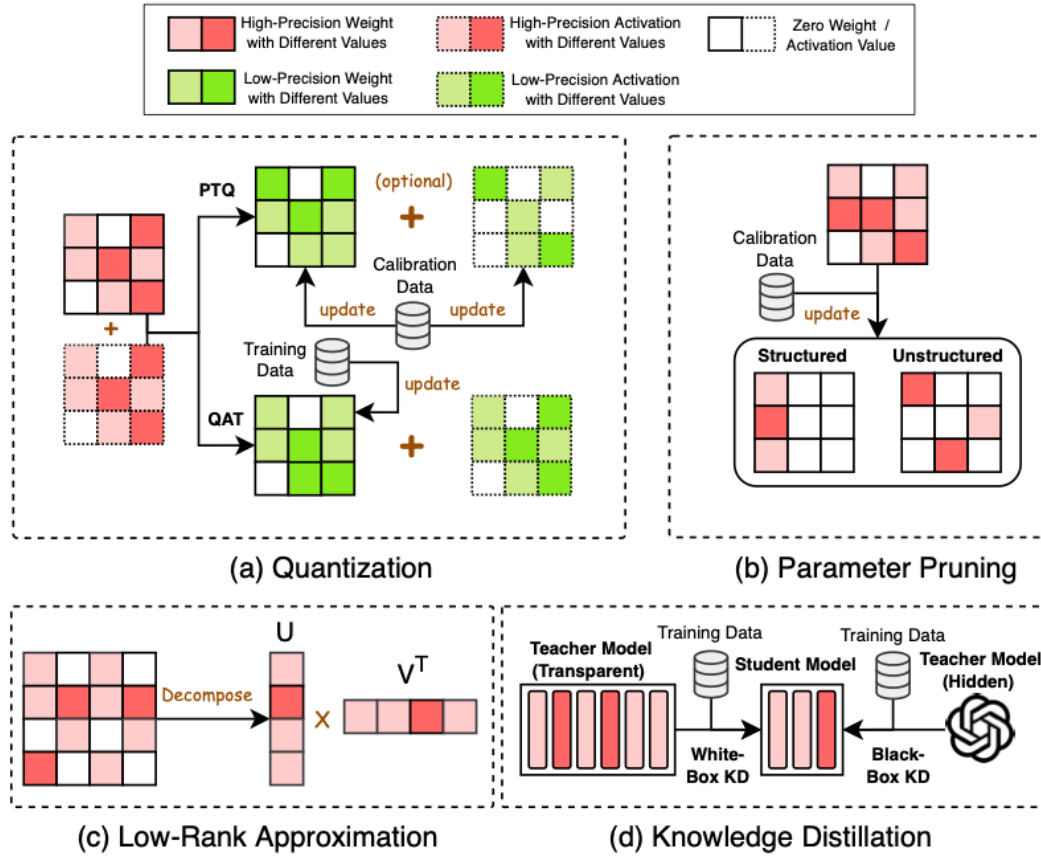
Dr. Kaoutar El Maghraoui

Knowledge Distillation

Spring 2026

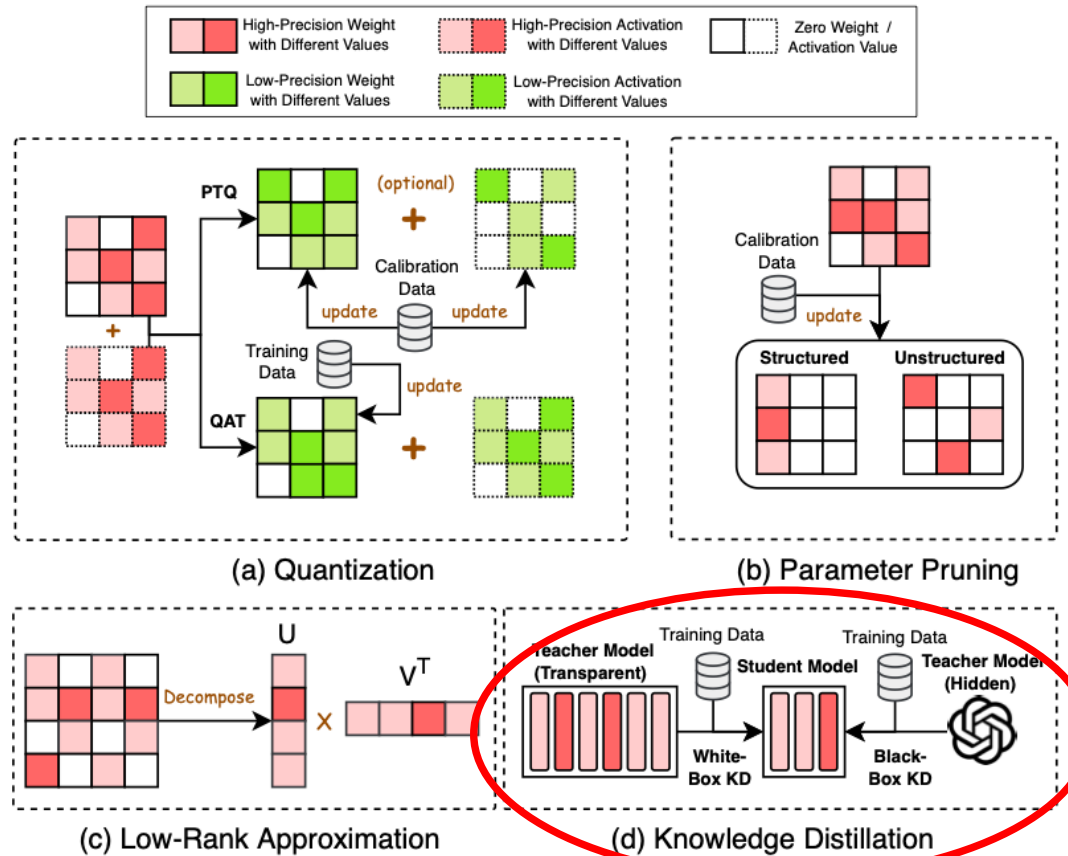


Illustration of Model Compression Techniques for LLMs



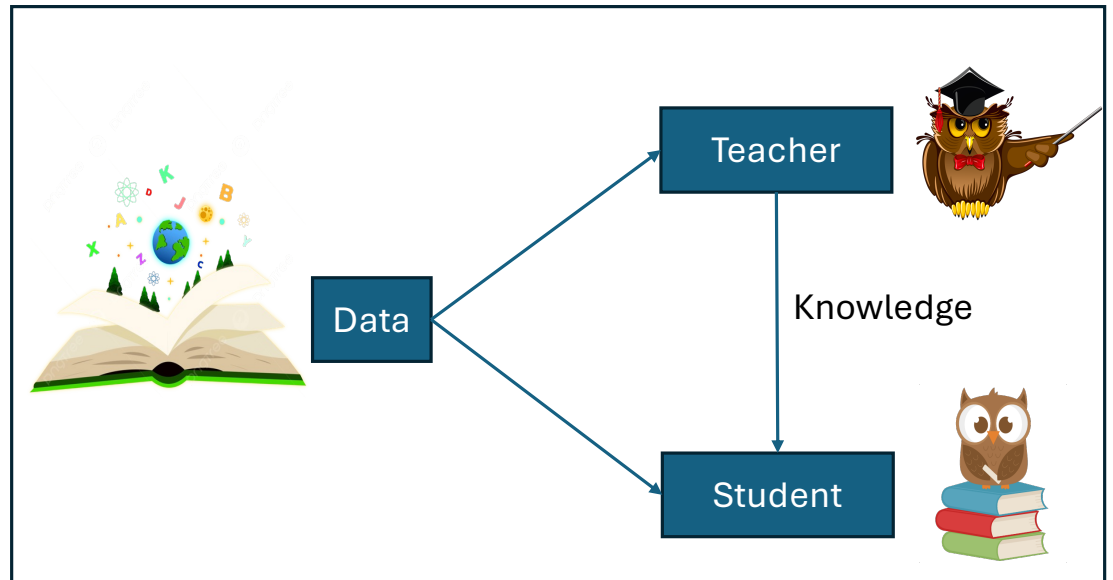
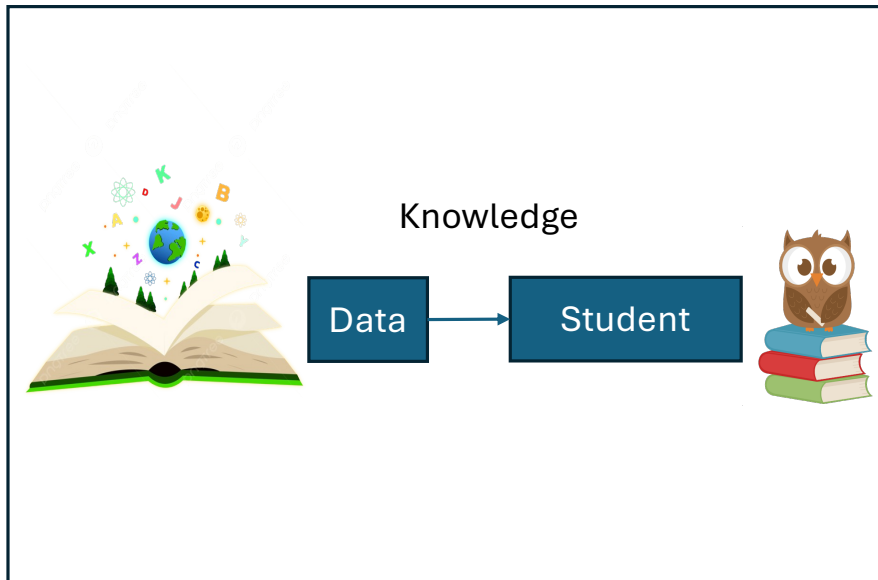
Efficient Large Language Models: A Survey
<https://arxiv.org/pdf/2312.103863.pdf>

Illustration of Model Compression Techniques for LLMs



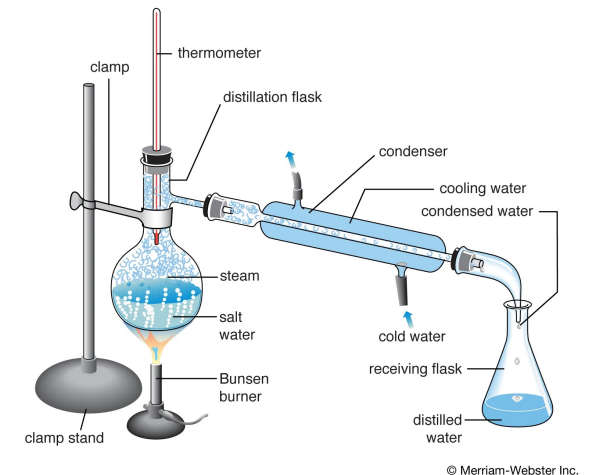
Efficient Large Language Models: A Survey
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What is Knowledge Distillation



What is Knowledge Distillation

- Distill knowledge from a large neural network to a small one
- Key Observations:
 - Large neural networks are easier to train
 - Small neural networks are easier to deploy
- The idea was first proposed by Caruana et al. in 2006
- The idea was generalized, and the term Knowledge Distillation was first coined by Hinton et al. in 2015



Water Distillation Analogy

Water Distillation Component	AI Model Distillation Equivalent
Salt water	Large, powerful teacher model (contains both useful knowledge and complexity)
Bunsen burner (heat source)	Training process (used to extract essential knowledge from the teacher)
Steam	Extracted knowledge/logits/representations from the teacher model
Condenser with cooling water	Distillation loss function — converts raw knowledge into learnable targets
Distilled water in receiving flask	Smaller, student model that retains purified knowledge with less complexity

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Distilling the Knowledge in a Neural Network

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Abstract

A very simple way to improve the performance of almost any machine learning algorithm is to train many different models on the same data and then to average their predictions [3]. Unfortunately, making predictions using a whole ensemble of models is cumbersome and may be too computationally expensive to allow deployment to a large number of users, especially if the individual models are large neural nets. Caruana and his collaborators [1] have shown that it is possible to compress the knowledge in an ensemble into a single model which is much easier to deploy and we develop this approach further using a different compression technique. We achieve some surprising results on MNIST and we show that we can significantly improve the acoustic model of a heavily used commercial system by distilling the knowledge in an ensemble of models into a single model. We also introduce a new type of ensemble composed of one or more full models and many specialist models which learn to distinguish fine-grained classes that the full models confuse. Unlike a mixture of experts, these specialist models can be trained rapidly and in parallel.

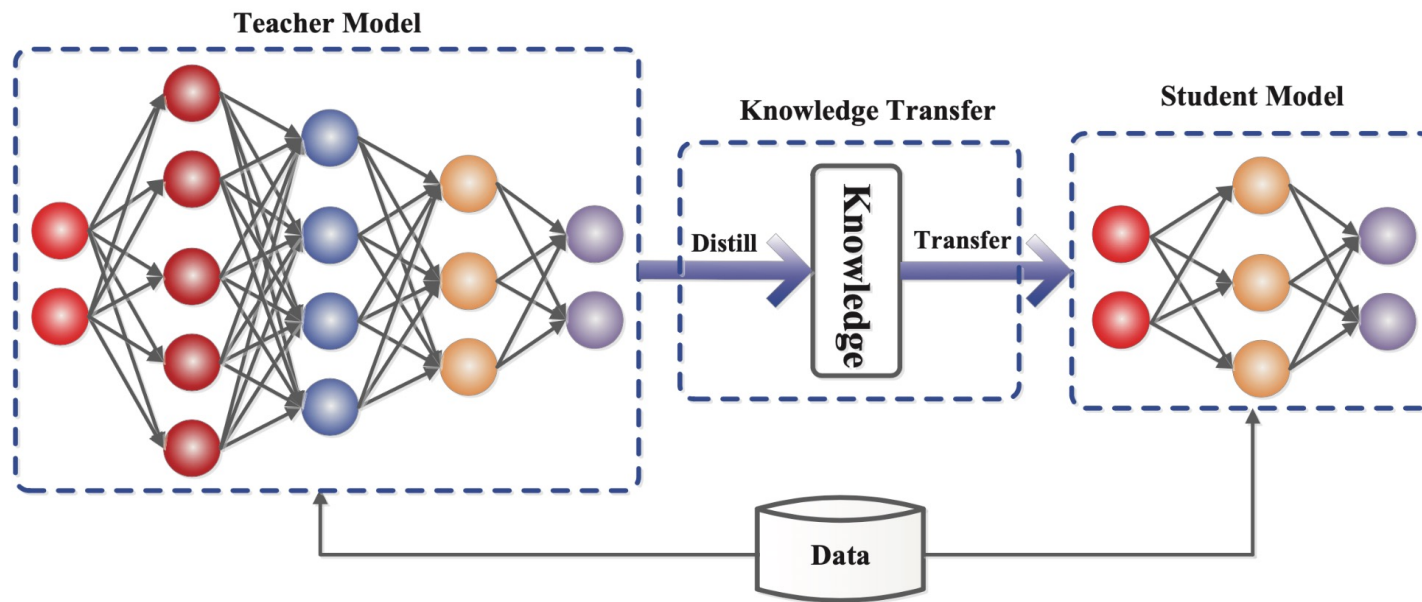
Popularization and Evolution

- After Hinton's 2015 paper, KD quickly gained adoption, especially for:
 - **Mobile and Edge AI** (e.g., TinyBERT, MobileNet with distillation).
 - **Large Language Models (LLMs)** (e.g., DistilBERT by Hugging Face).
 - **Computer Vision** (ResNet distillation, EfficientNet-TF Lite).
- *DistilBERT: A distilled version of BERT* (Sanh et al., 2019)
 - Hugging Face introduced DistilBERT, which used KD to reduce BERT's size by **40%** while retaining **97%** of its language understanding.
- KD has also expanded into **task-specific** distillation (e.g., question answering), **multi-teacher** settings, **intermediate feature distillation**, and even **self-distillation**.

Year	Event
2006	Caruana et al. propose model compression using soft labels from ensembles.
2015	Hinton, Vinyals, Dean formalize Knowledge Distillation .
2019	Hugging Face releases DistilBERT , popularizing KD in NLP.
2020+	KD extended to vision transformers, multi-modal models, LLMs, and cross-task knowledge transfer.

A More Formal Definition

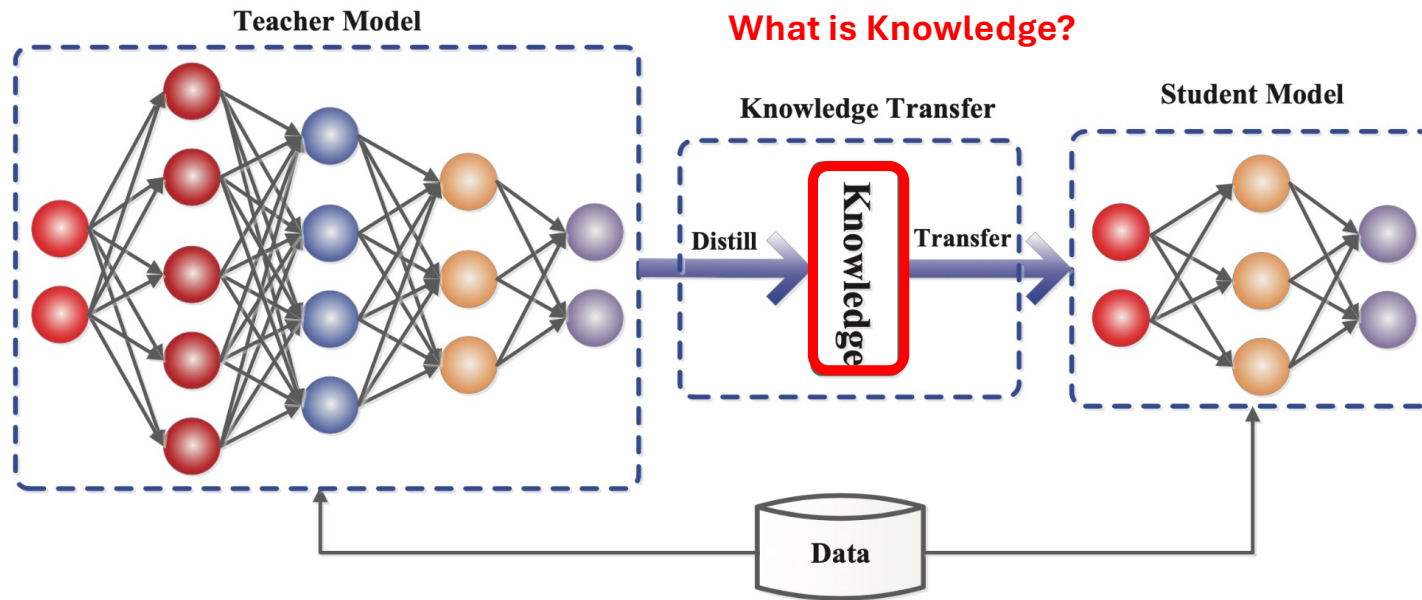
Knowledge distillation is a machine learning technique in which a smaller model, often referred to as a student model, is trained to replicate the behavior or predictions of a larger, more complex model, known as the teacher model.



The teacher-student framework for knowledge distillation | Source: [Arxiv](#)

A More Formal Definition

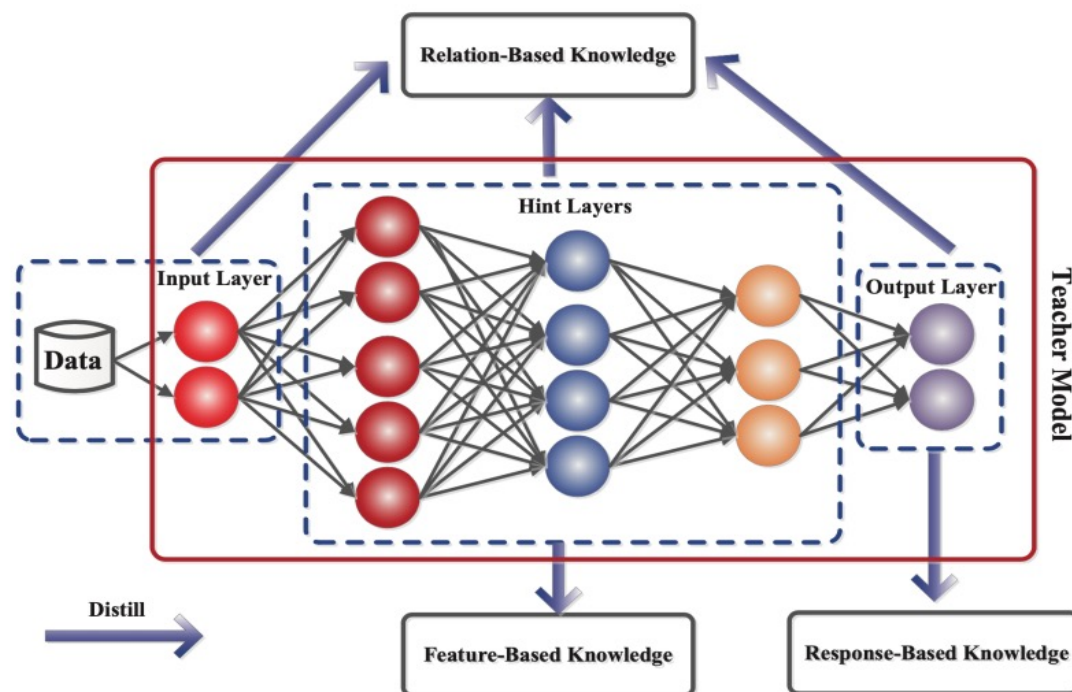
Knowledge distillation is a machine learning technique in which a smaller model, often referred to as a student model, is trained to replicate the behavior or predictions of a larger, more complex model, known as the teacher model.



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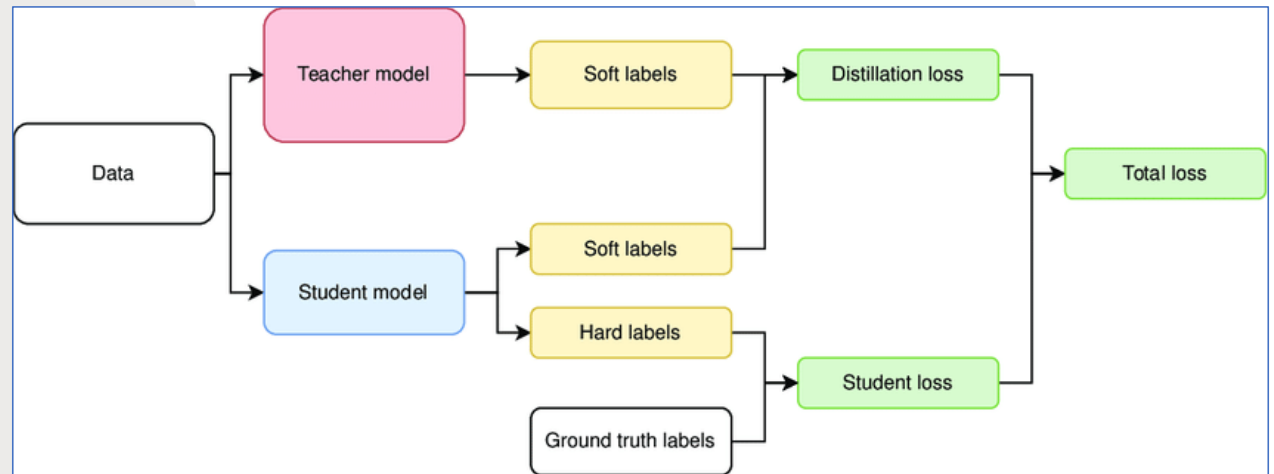
What is Knowledge?

- In a neural network (DNN), "knowledge" refers to the learned weights and biases.
- A rich diversity exists in the sources of knowledge within a large DNN
- Knowledge distillation commonly uses logits as the source of teacher knowledge, while other approaches focus on weights, activations of intermediate layers, or other relevant parameters.
- Various forms of knowledge can be categorized into three types:
 - Response-based knowledge
 - Feature-based knowledge
 - Relation-based knowledge



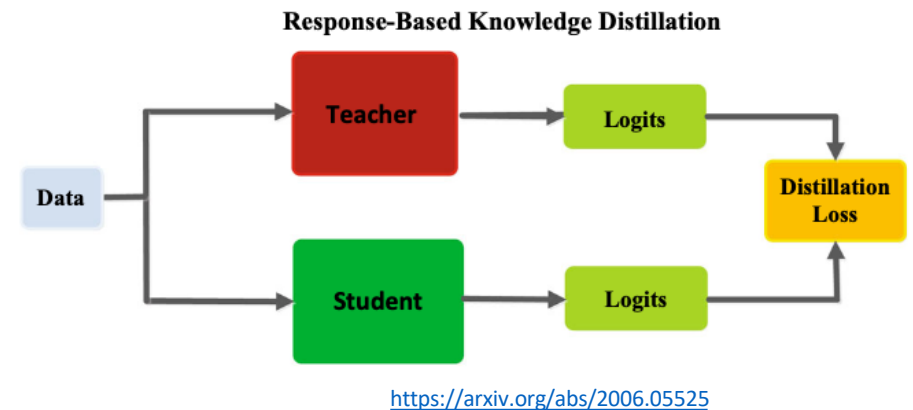
The different kinds of knowledge in a teacher model | Source: [Arxiv](#)

Response- Based Knowledge



Response-based Knowledge

- Refers to the neural response of the last output layer in the teacher model.
- Rational: Mimic the final prediction of the teacher model
- Proposed by *Geoffrey Hinton, Oriol Vinyals, and Jeff Dean (2015)*.
“Distilling the knowledge in a neural network”



Key Concepts

- Soft Label
- Modified Softmax with Temperature
- Distillation Loss

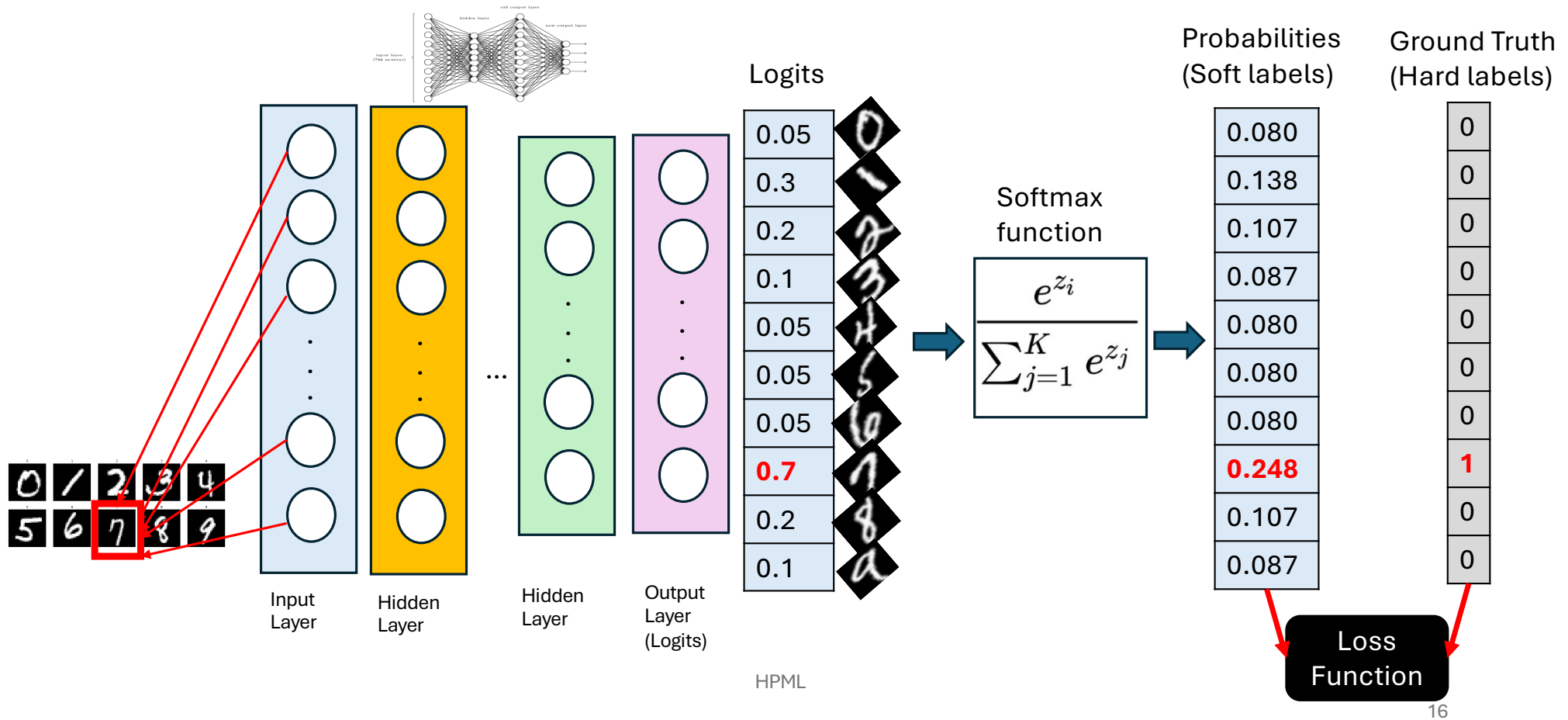


Key Concepts

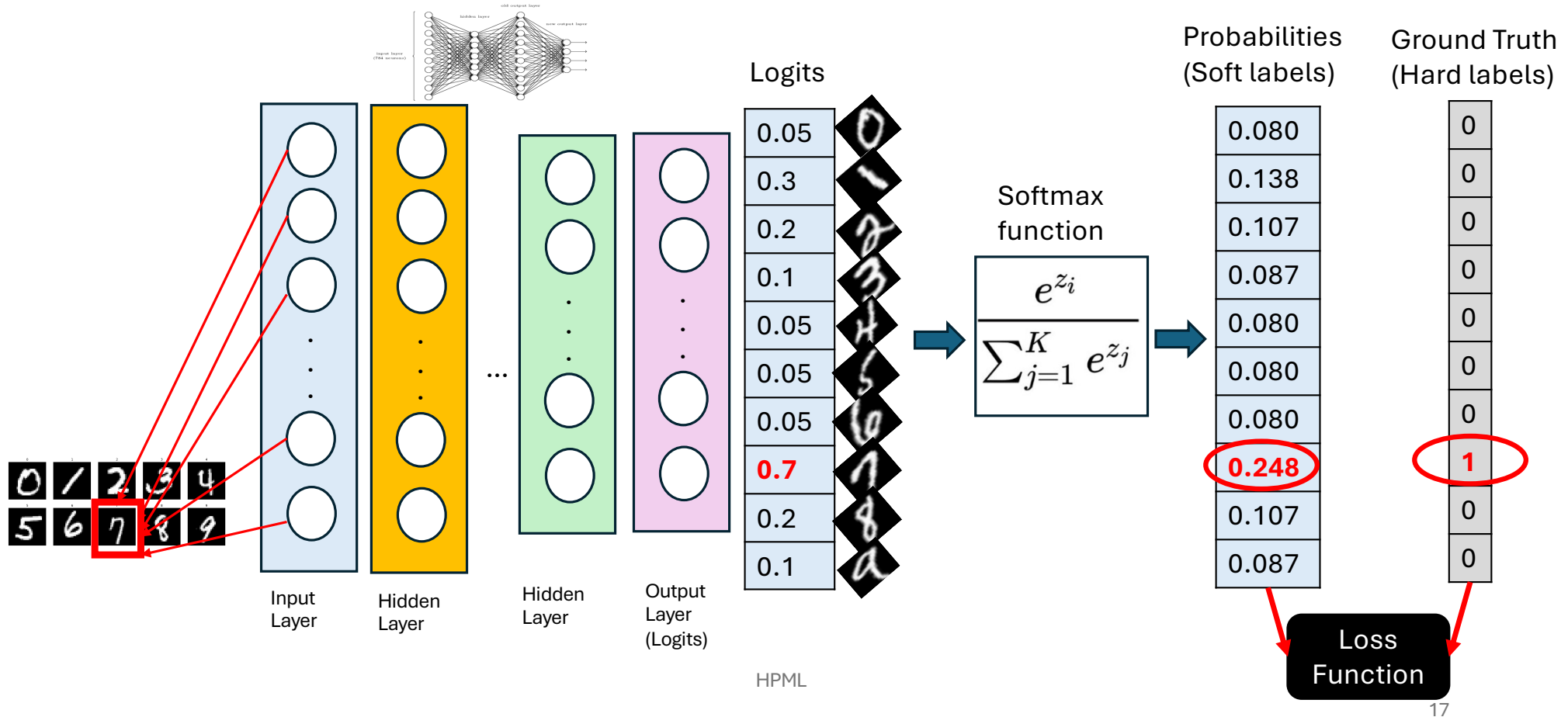
- Soft Label
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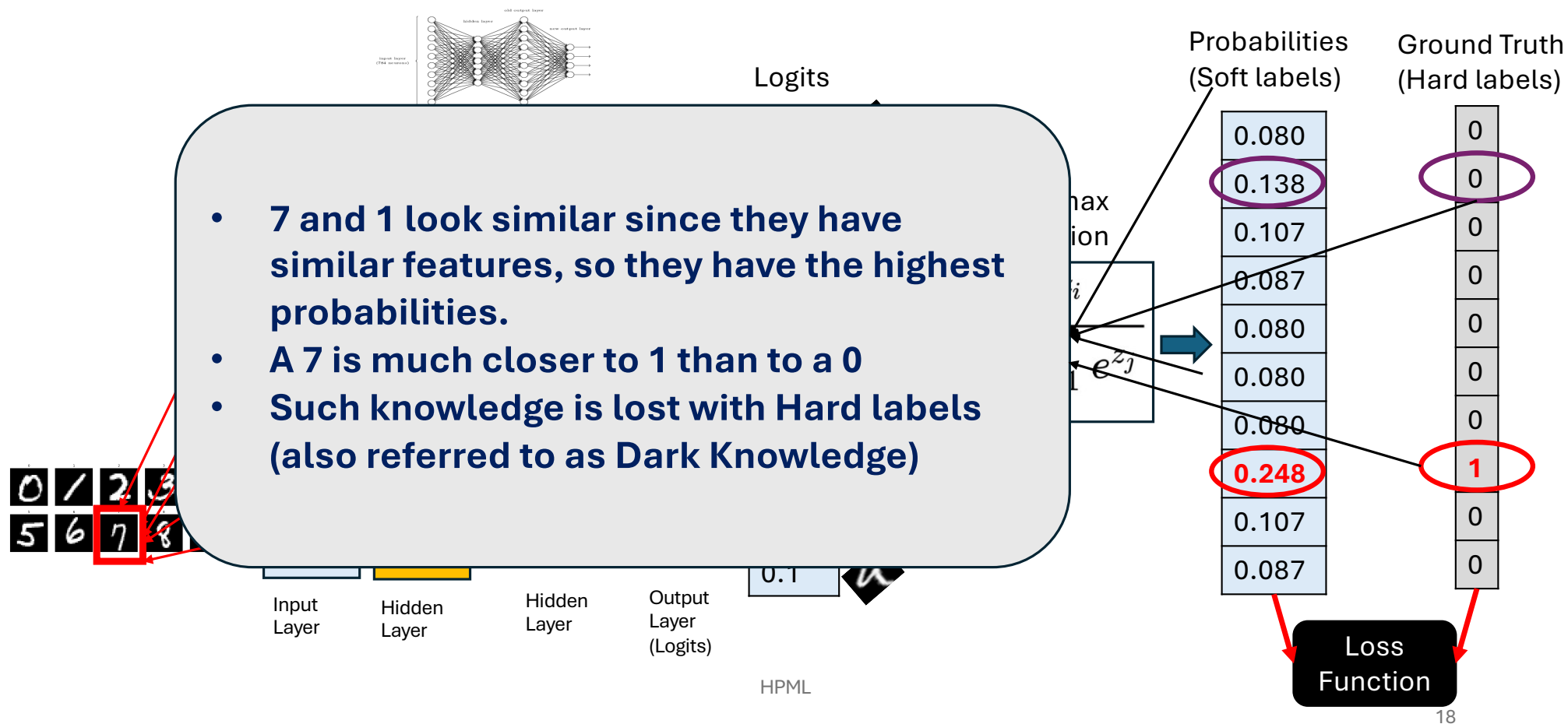
Hard vs. Soft Labels



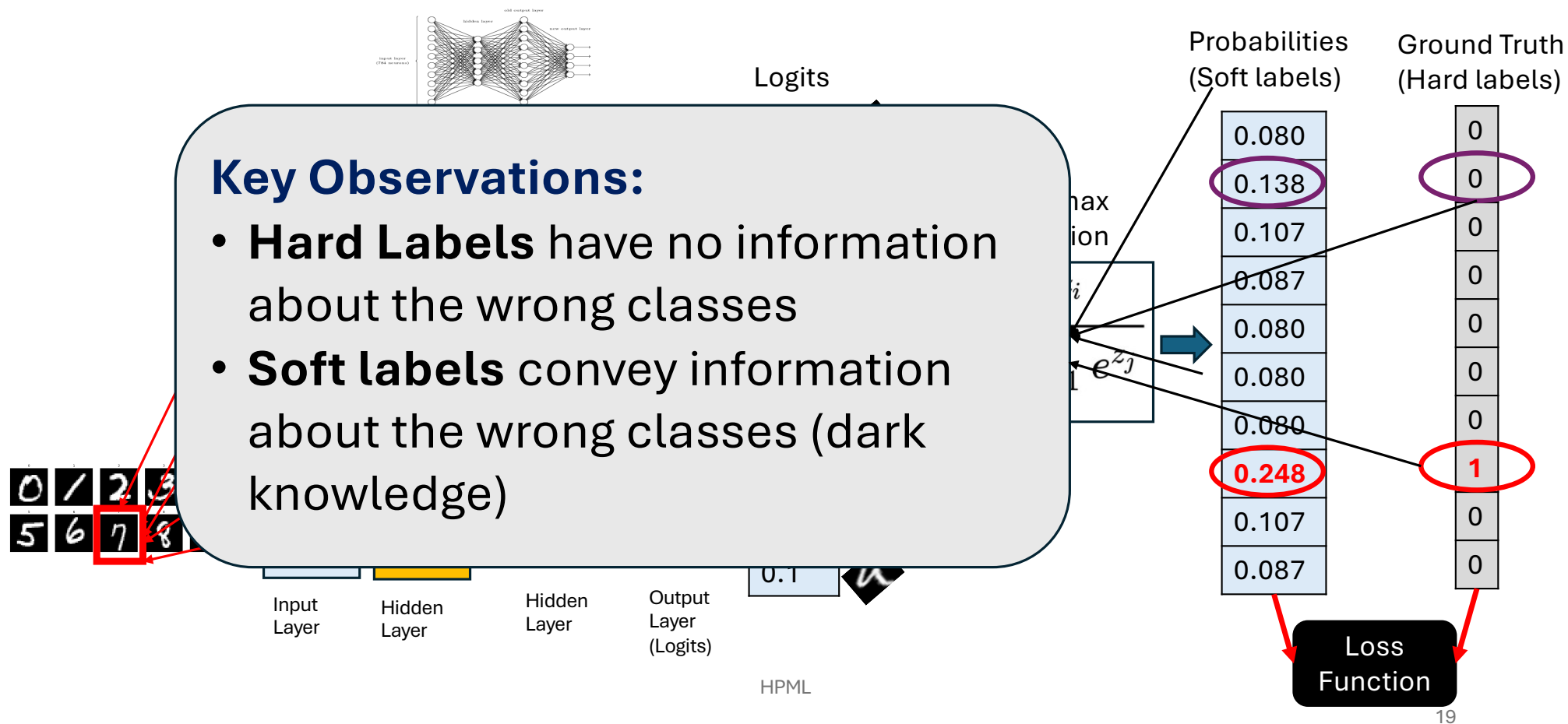
Hard vs. Soft Labels (Targets)



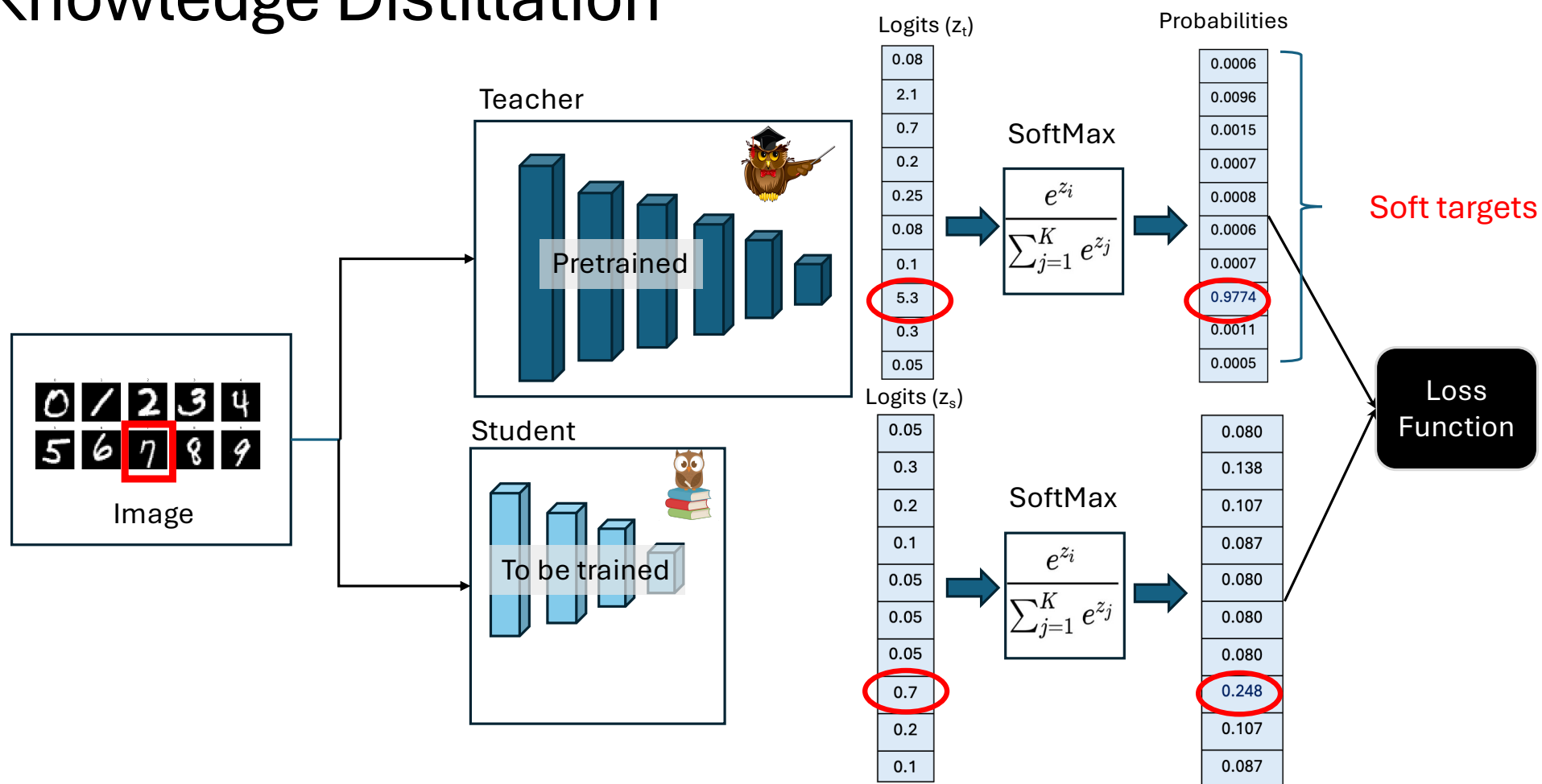
Hard vs. Soft Labels



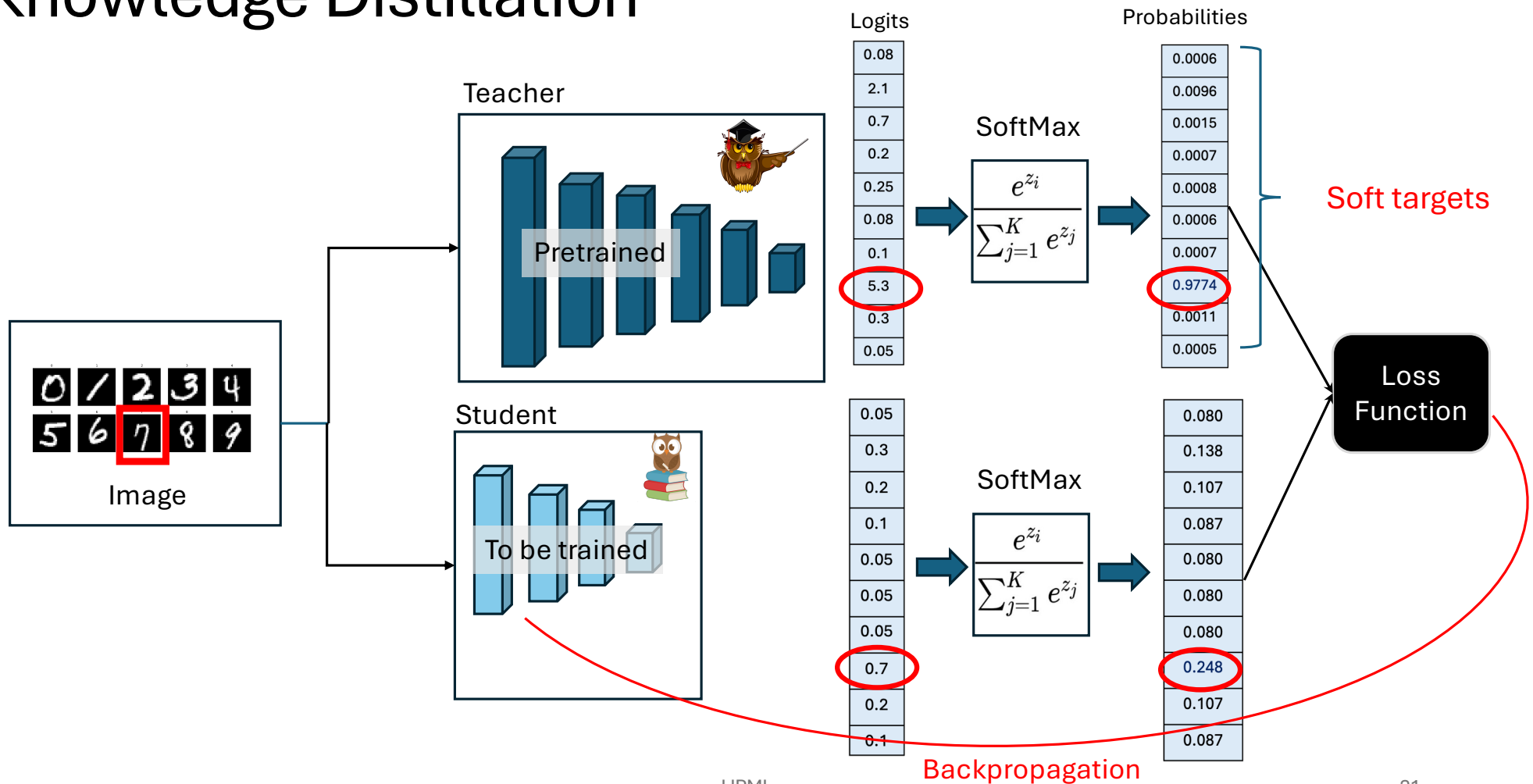
Hard vs. Soft Labels



Knowledge Distillation



Knowledge Distillation



Summary

- A handwritten 7 digit is much closer to a handwritten 1 than a handwritten 8.
- Training on probabilities as “soft targets” is much better for knowledge distillation:
 - More informative.
 - Less variance in gradients leads to faster training since the learning rate can be increased.

Key Concepts

- Soft Label
- Modified Softmax with Temperature
- Distillation Loss



Modified Softmax with Temperature

$$q_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

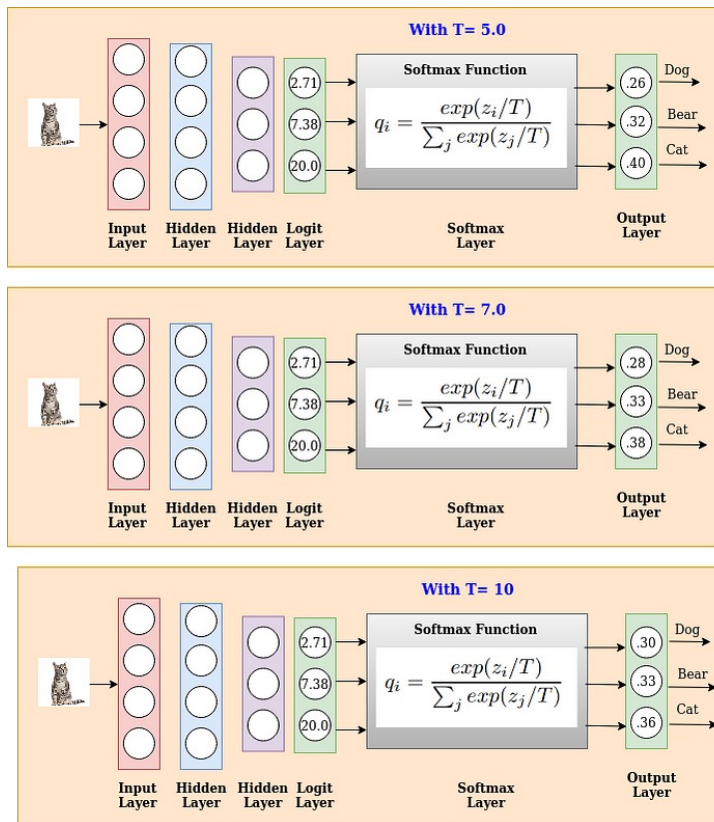
Normal Softmax

$$q_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

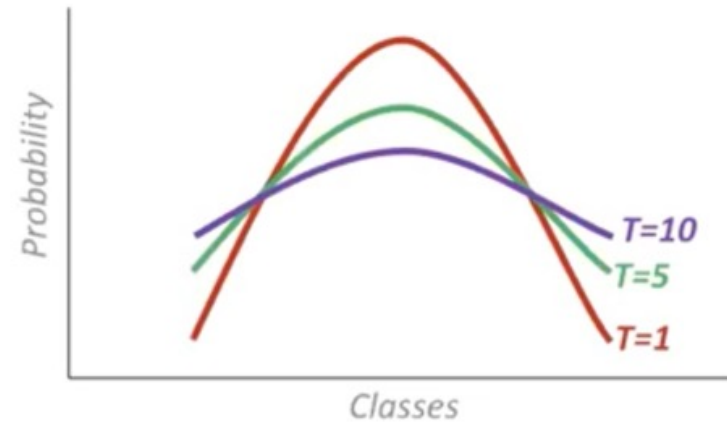
Softmax with Temperature

- Z_i are the logits of the neural network
- T called the temperature, is a new hyperparameter
- We are increasing the entropy in the target labels

Example



[Example source](#)



- When $T > 1$, the temperature smooths out the output distribution and carries more information about all the target types.
- This information is squeezed when we use the normal softmax function.
- **Soft labels** are the results of dividing the logits with the temperature in the softmax

<https://colab.research.google.com/drive/121utlMBx2WedV0dLo6gwgPot3W4hEnVf>

Key Concepts

- Soft Label
- Modified Softmax with Temperature
- Distillation Loss



Distillation Loss

- The soft labels between Teacher and Student need to be optimized using **KL(Kullback-Leibler) Divergence loss**.
- KL divergence:
 - Non-symmetric metric
 - Measures relative entropy or difference in information between distributions
 - Indicates the distance between two data distributions
 - Reflects dissimilarity between distributions

KL Divergence Definition

Kullback–Leibler divergence (KL divergence), D_{KL} ³, is a measure of how one probability distribution is different from a second

For distributions P and Q of a discrete random variable, the KL divergence from P to Q is:

$$D_{KL}(p(x) || q(x)) = \sum_{x \in X} p(x) \ln \frac{p(x)}{q(x)}$$

For distributions P and Q of a continuous random variable, the KL divergence from P to Q is:

$$D_{KL}(p(x) || q(x)) = \int_{-\infty}^{\infty} p(x) \ln \frac{p(x)}{q(x)} dx$$

- $D_{KL}(P||Q)$ is the KL divergence between distributions P and Q .
- $P(i)$ and $Q(i)$ are the probabilities of event i in distributions P and Q , respectively.

Reference: Solomon Kullback and Richard A Leibler (1951). "On information and sufficiency". In: vol. 22. JSTOR, pp. 79–86.

Distillation Loss

In the context of knowledge distillation, the distillation loss L_{distill} using the KL divergence can be expressed as the average KL divergence between the soft targets and the student predictions over the entire dataset:

$$L_{\text{distill}} = \frac{1}{N} \sum_i D_{\text{KL}}(P_{\text{soft}}(i) \parallel Q_{\text{student}}(i))$$

Where:

- N is the total number of samples in the dataset.
- $P_{\text{soft}}(i)$ is the probability distribution of the soft targets (teacher model) for event i .
- $Q_{\text{student}}(i)$ is the probability distribution of the student model predictions for event i .

"event i " refers to the i^{th} **sample** or **data point** in your dataset.

This distillation loss encourages the student model to produce similar output distributions to the soft targets, transferring knowledge from the teacher to the student.

Student Loss

In knowledge distillation, the student loss refers to the traditional loss function used to train the student model, typically involving the comparison between the predictions of the student model and the ground truth labels.

$$L_{\text{student}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$$

Where:

- N is the total number of samples.
- C is the number of classes.
- $y_{i,c}$ is a binary indicator function that equals 1 if the sample i belongs to class c and 0 otherwise.
- $p_{i,c}$ is the predicted probability that sample i belongs to class c .

Total Loss

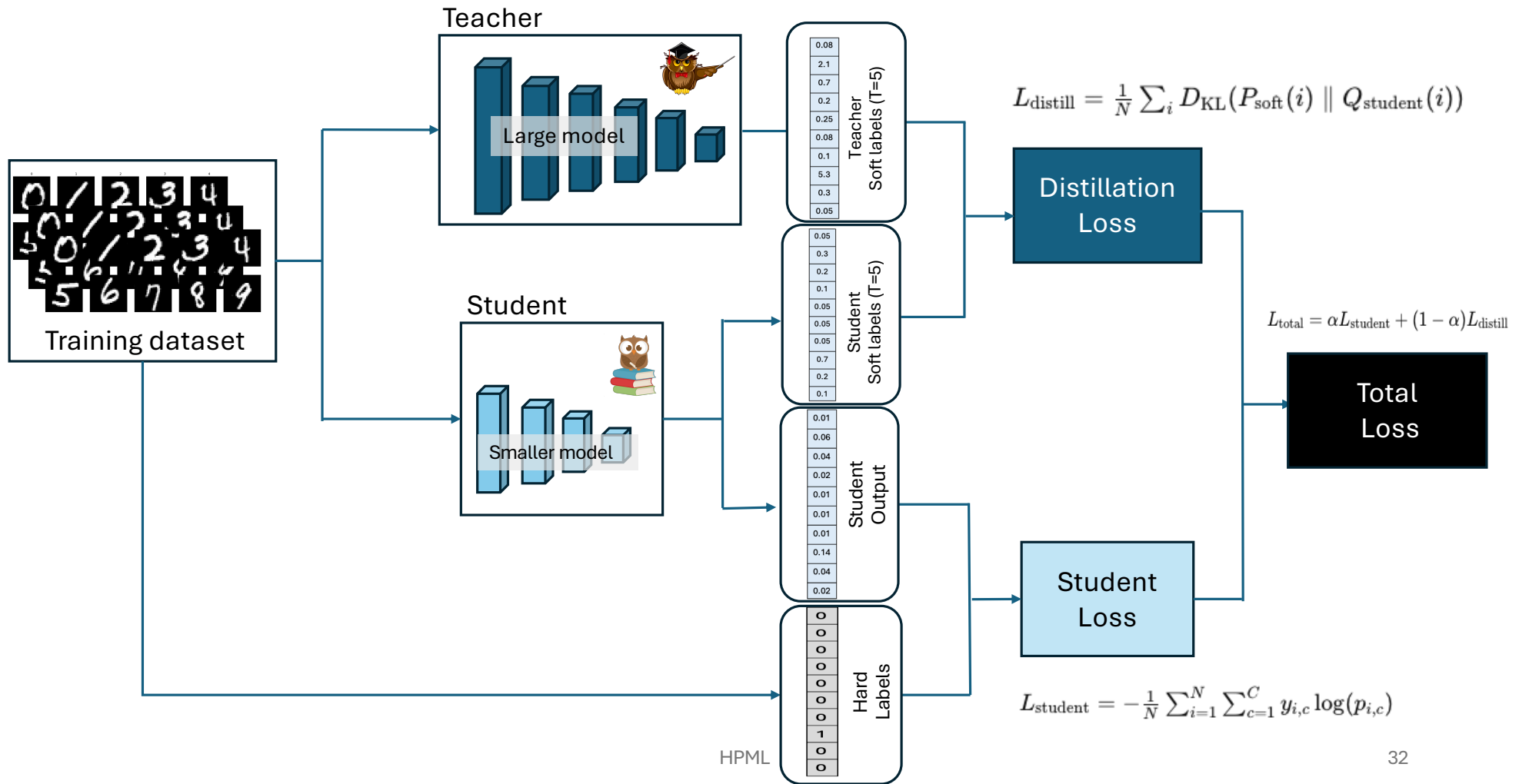
The total loss in knowledge distillation is a combination of the student loss and the distillation loss. The total loss L_{total} can be expressed as a weighted sum of the student loss L_{student} and the distillation loss L_{distill} :

$$L_{\text{total}} = \alpha L_{\text{student}} + (1 - \alpha) L_{\text{distill}}$$

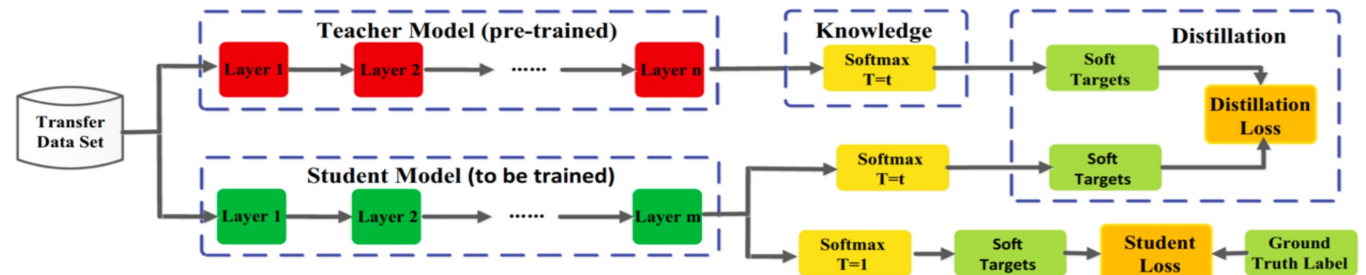
Where:

- L_{student} is the loss function used to train the student model on the task-specific labels.
- L_{distill} is the distillation loss, which measures the difference between the output distributions of the teacher and student models.
- α is a hyperparameter that controls the balance between the two components of the loss. It typically ranges between 0 and 1, with higher values indicating more emphasis on the student loss and lower values indicating more emphasis on the distillation loss.

Loss Function



Formulation of Response-Based Knowledge (1/2)



Geoffrey Hinton, Oriol Vinyals, and Jeff Dean (2015). "Distilling the knowledge in a neural network"

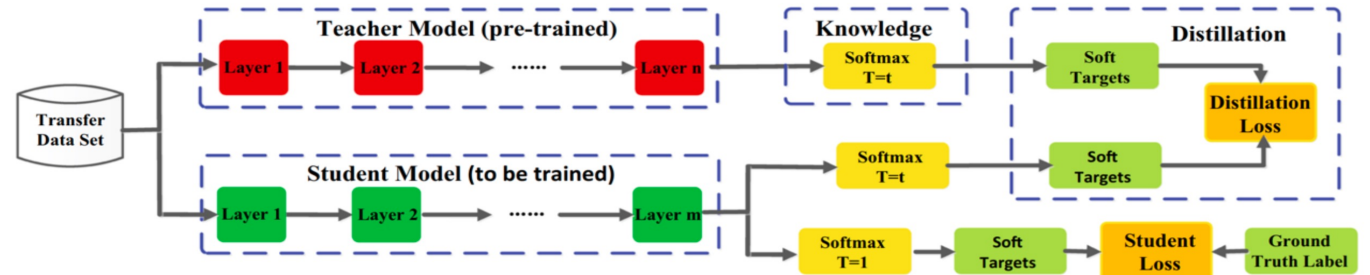
Formulation

Given a vector of logits z as the outputs of the last fully connected layer of a deep model, the distillation loss for response-based knowledge can be formulated as:

$$L_{ResD}(z_t, z_s) = \mathcal{L}_R(z_t, z_s)$$

where $\mathcal{L}_R(\cdot)$ indicates the divergence loss of logits, and z_t and z_s are logits of teacher and student respectively.

Formulation of Response-Based Knowledge (1/2)



Geoffrey Hinton, Oriol Vinyals, and Jeff Dean (2015). "Distilling the knowledge in a neural network"

- Soft targets are the most popular response-based knowledge.
- Soft targets are the probabilities that the input belongs to the classes and can be estimated by a softmax function as:

$$p(z_i, T) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

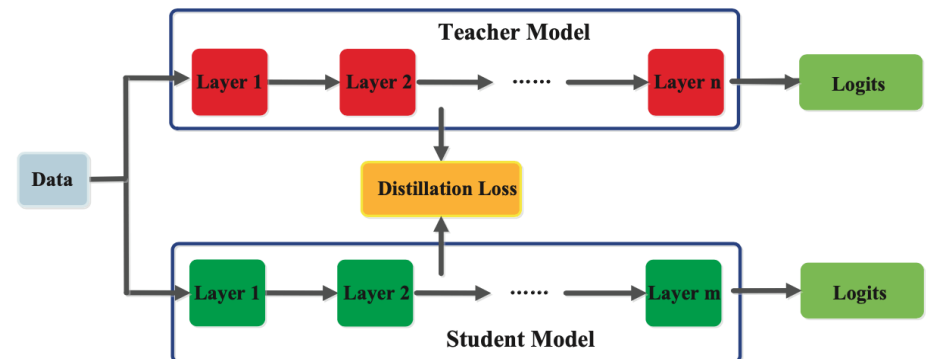
where z_i is the logit for the i_{th} class, and a temperature factor T is introduced to control the importance of each soft target. Accordingly, the distillation loss for soft logits can be rewritten as

$$L_{ResD}(p(z_t, T), p(z_s, T)) = \mathcal{L}_R(p(z_t, T), p(z_s, T))$$

What are the limitations of Response-based Knowledge?

- Response-based knowledge distillation typically relies on the output of the last layer, such as soft targets.
- It often overlooks intermediate-level supervision provided by the teacher model, which is crucial for representation learning in deep neural networks.
- The reliance on soft logits, representing class probability distribution, limits response-based knowledge distillation to supervised learning tasks.
- Research suggests that incorporating intermediate-level supervision is vital for effectively distilling knowledge from teacher-to-student models.

Feature-Based Knowledge



Feature-based Knowledge



Definition: Feature-based knowledge distillation involves training the student model to mimic the internal representations or features learned by the teacher model.



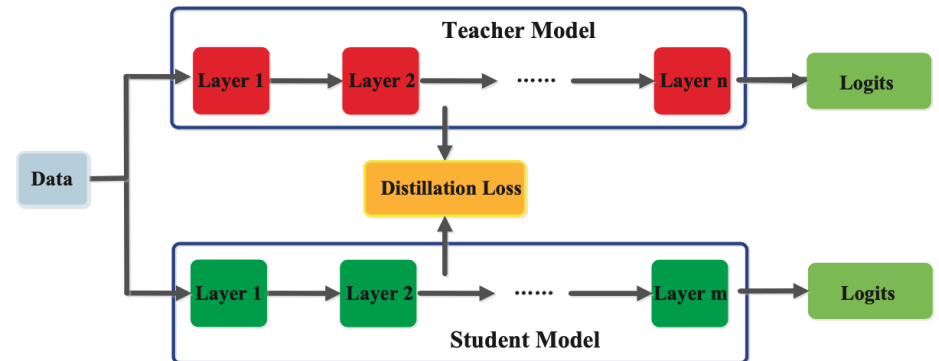
Process: The teacher model's internal representations are extracted from intermediate layers and used as targets for the student model.



Goal: The process aims to minimize the distance between the features of the teacher and student model using loss functions like mean squared error or Kullback-Leibler divergence.

Distillation Process

- Teacher Model Training: Learns task-specific features from training data.
- Student Model Training: Mimics teacher's features by minimizing feature distance.
- Loss Function: Measures distance between teacher and student model's representations.



Formulation of Feature-based Knowledge Loss Function

Formulation

Generally, the distillation loss for feature-based knowledge transfer can be formulated as

$$L_{FeaD}(f_t(x), f_s(x)) = \mathcal{L}_F(\phi_t(f_t(x)), \phi_s(f_s(x)))$$

where $f_t(x)$ and $f_s(x)$ are the feature maps of the intermediate layers of teacher and student models, respectively. The transformation functions, $\phi_t(f_t(x))$ and $\phi_s(f_s(x))$, are usually applied when the feature maps of teacher and student models are not in the same shape. $\mathcal{L}_F(\cdot)$ indicates the similarity function used to match the feature maps of teacher and student models. $\mathcal{L}_F(\cdot)$ can be $\mathcal{L}_2(\cdot)$, $\mathcal{L}_1(\cdot)$, $\mathcal{L}_{CE}(\cdot)$ and *etc.*

$L_2(\cdot)$, $L_1(\cdot)$, $L_{CE}(\cdot)$ and $L_{MMD}(\cdot)$ indicate l2-norm distance, l1-norm distance, cross-entropy loss and maximum mean discrepancy loss, respectively

Example: FitNets - Hints for Thin Deep Nets

Published as a conference paper at ICLR 2015

FITNETS: HINTS FOR THIN DEEP NETS

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²Université de Montréal, Montréal, Québec, Canada. [†]CIFAR Senior Fellow.

³École Polytechnique de Montréal, Montréal, Québec, Canada.

⁴Centre de Visió per Computador, Bellaterra, Spain.

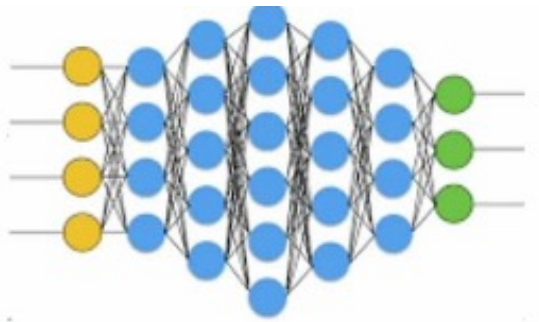
ABSTRACT

While depth tends to improve network performances, it also makes gradient-based training more difficult since deeper networks tend to be more non-linear. The recently proposed knowledge distillation approach is aimed at obtaining small and fast-to-execute models, and it has shown that a student network could imitate the soft output of a larger teacher network or ensemble of networks. In this paper, we extend this idea to allow the training of a student that is deeper and thinner than the teacher, using not only the outputs but also the intermediate representations learned by the teacher as hints to improve the training process and final performance of the student. Because the student intermediate hidden layer will generally be smaller than the teacher's intermediate hidden layer, additional parameters are introduced to map the student hidden layer to the prediction of the teacher hidden layer. This allows one to train deeper students that can generalize better or run faster, a trade-off that is controlled by the chosen student capacity. For example, on CIFAR-10, a deep student network with almost 10.4 times less parameters outperforms a larger, state-of-the-art teacher network.

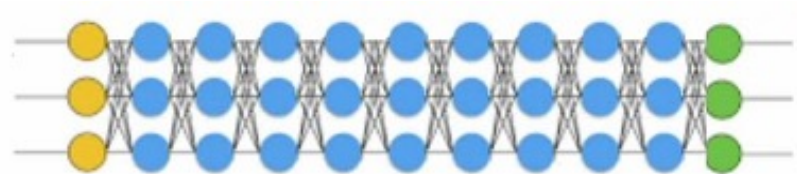
<https://arxiv.org/pdf/1412.6550>

- Motivation: prior knowledge distillation work did not consider the depth of the student model
- This paper extends knowledge distillation to train a deeper and thinner student using teacher outputs and intermediate representations as hints.
- Depth Improvement vs. Training Complexity:
 - Deeper networks enhance performance but pose challenges for gradient-based training due to increased non-linearity.

FitNets

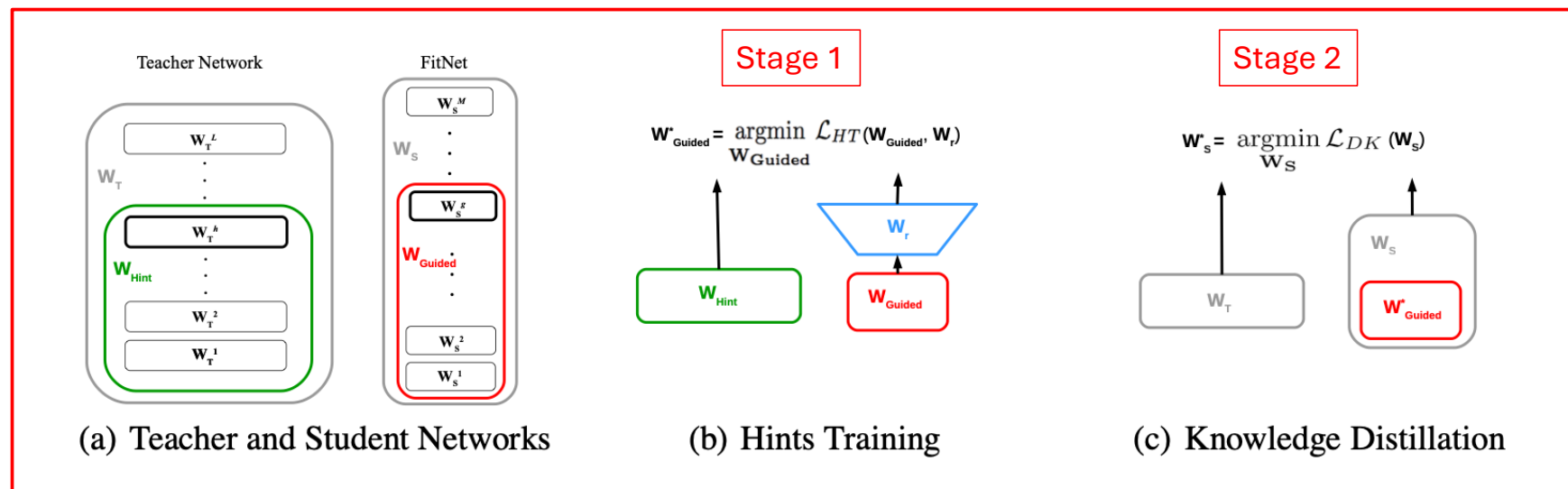


- **Top Performing Models**
 - Deep and wide
 - Large number of parameters



- **Fitnets (Focus on Depth)**
 - Deeper and thinner and wide
 - Lesser or equal parameters than the teacher
 - Hint-based stage wise training

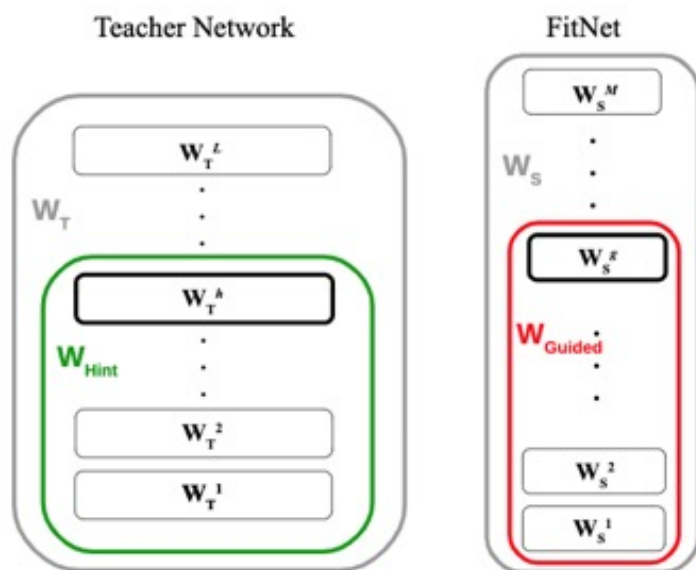
FitNets Staged Knowledge Distillation Training



Training a student network using hints

<https://arxiv.org/pdf/1412.6550>

Stage 1: Training the guided subnetwork

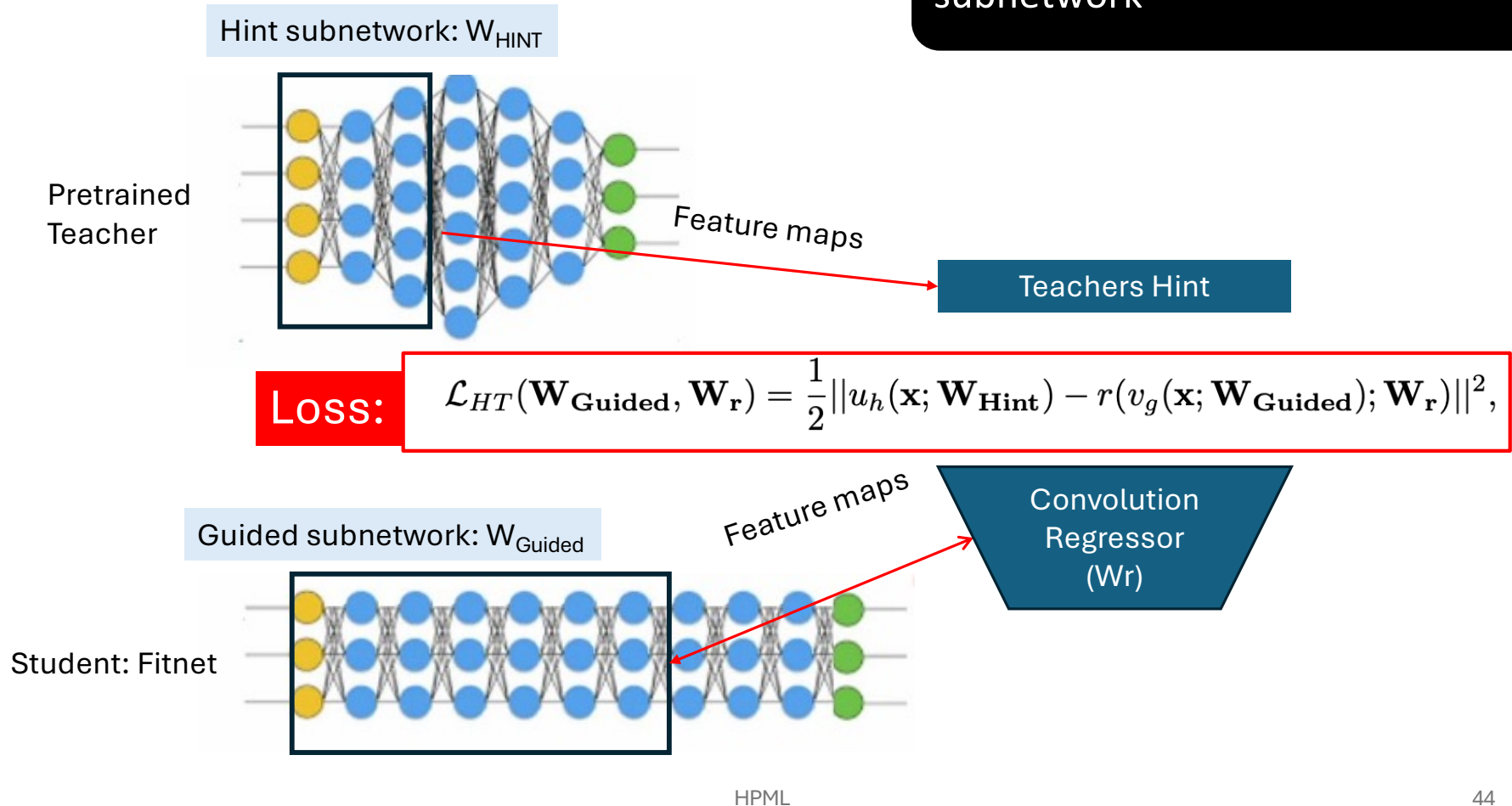


“**Hint**” layer: The layer from the teacher whose output a student should learn to predict

“**Guided**” layer: The layer from the student network that learns

Stage 1

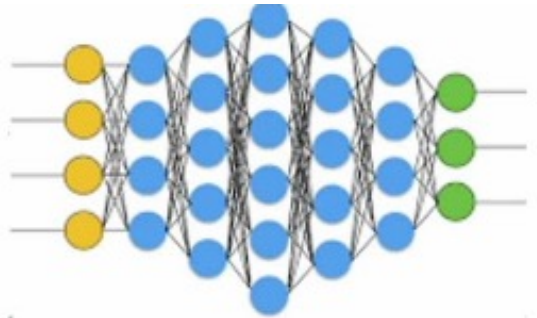
Stage 1: Training of the guided subnetwork



Stage 2

Stage 2: Training of entire student model using the regular KD process

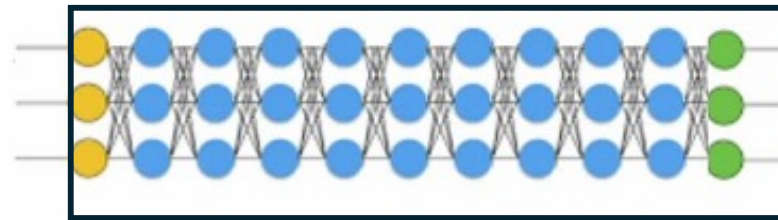
Pretrained Teacher



Loss:

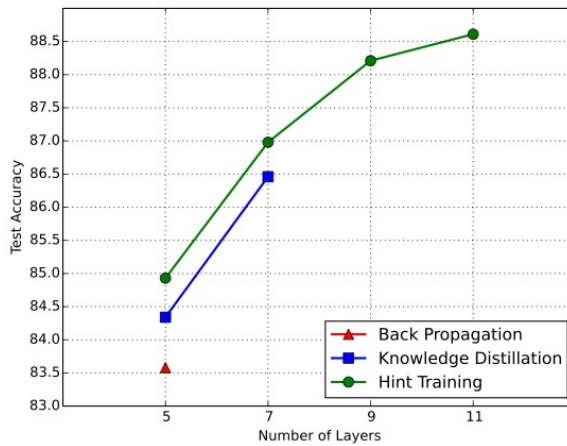
$$\mathcal{L}_{KD}(\mathbf{W}_S) = \mathcal{H}(\mathbf{y}_{\text{true}}, P_S) + \lambda \mathcal{H}(P_T^\tau, P_S^\tau),$$

Student: Fitnet

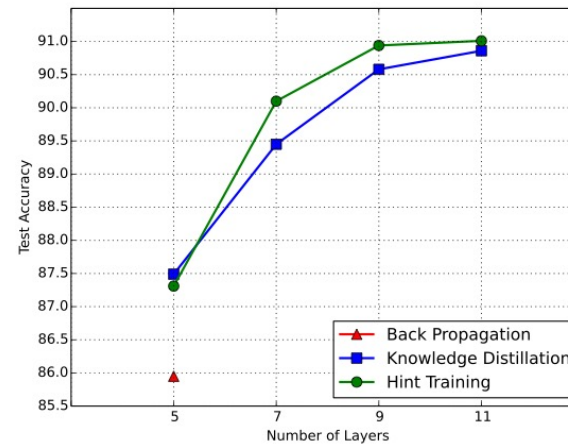


HPML

Experimental Evaluation: Effect of Hint-based Training



(a) 30M Multiplications



(b) 107M Multiplications

Comparison of Standard Back-Propagation, Knowledge Distillation, and Hint-based Training on CIFAR-10.

Experimental Evaluation: Effect of Hint-based Training

Network	# layers	# params	# mult	Acc	Speed-up	Compression rate
Teacher	5	~9M	~725M	90.18%	1	1
FitNet 1	11	~250K	~30M	89.01%	13.36	36
FitNet 2	11	~862K	~108M	91.06%	4.64	10.44
FitNet 3	13	~1.6M	~392M	91.10%	1.37	5.62
FitNet 4	19	~2.5M	~382M	91.61%	1.52	3.60

Accuracy/Speed Trade-off on CIFAR-10



Fitnets Key Takeaways

Using Hint-based training, we are able to train deeper models compared with standard back propagation

Given a fixed capacity, deeper models perform better than shallower ones

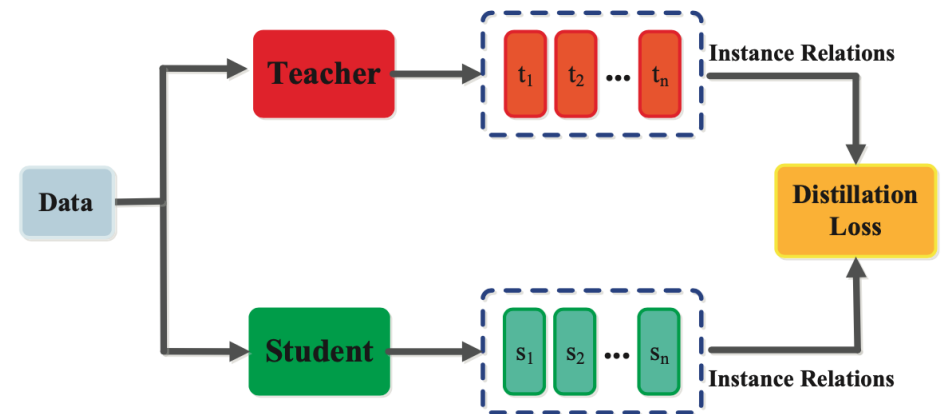
Example: Feature-based KD using Attention Transfer

This work proposed a feature-based knowledge distillation using the attention mechanism. Specifically, the student network learns attention information from the teacher network.

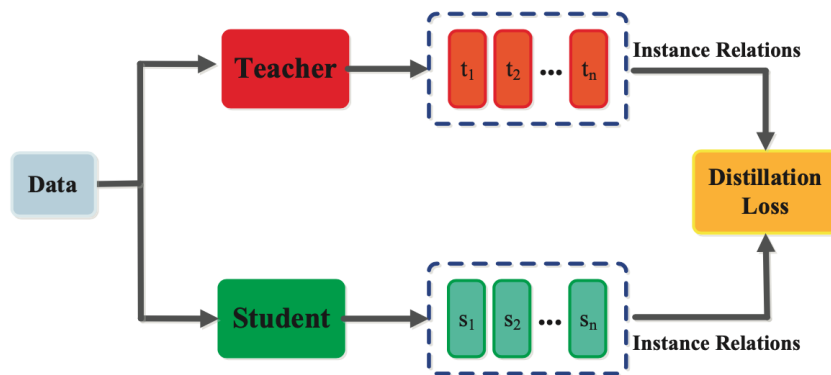


Sergey Zagoruyko and Nikos Komodakis (2016). "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer". In: *arXiv preprint arXiv:1612.03928*

Relation-Based Knowledge



Relation-based Knowledge



- Captures relationships between feature maps to train a student model.
- Relationships can be modeled as:
 - Correlation between feature maps
 - Graphs
 - Similarity matrix
 - Feature embeddings
 - Probabilistic distributions based on feature representations.

Relation-based Loss Function

Formulation

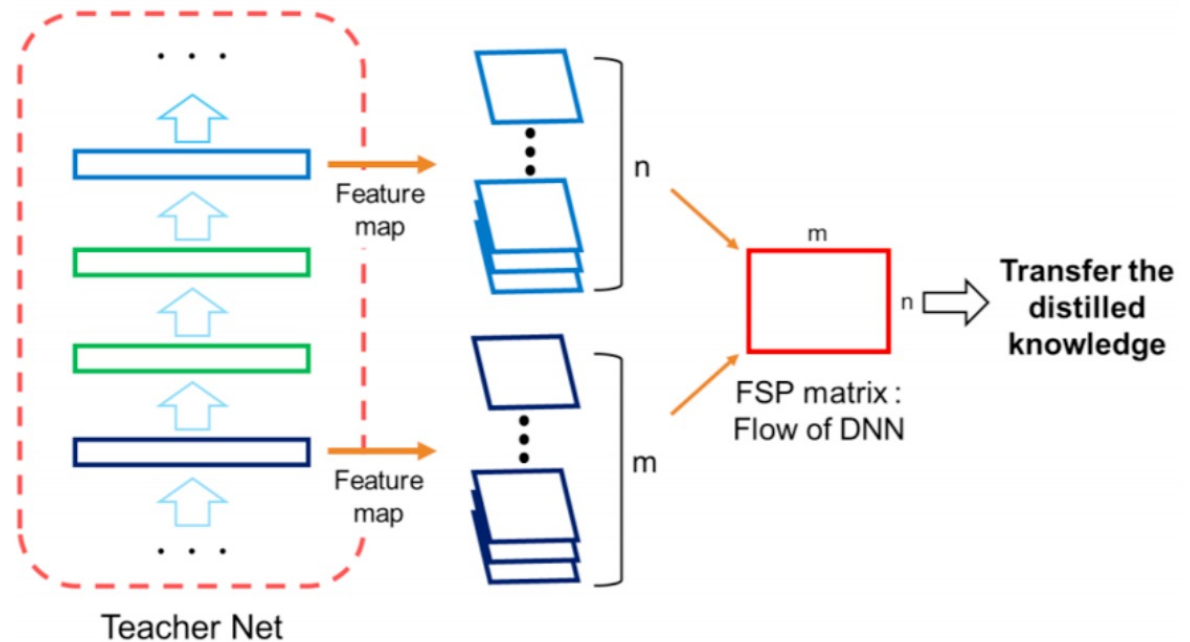
In general, the distillation loss of relation-based knowledge based on the relations of feature maps can be formulated as

$$L_{RelD}(f_t, f_s) = \mathcal{L}_R(\Phi_t(\hat{f}_t, \check{f}_t), \Phi_s(\hat{f}_s, \check{f}_s))$$

where f_t and f_s are the feature maps of teacher and student models, respectively. Pairs of feature maps are chosen from the teacher model, $\hat{f}_t$ and $\check{f}_t$, and from the student model, $\hat{f}_s$ and $\check{f}_s$. $\Phi_t(\cdot)$ and $\Phi_s(\cdot)$ are the similarity functions for pairs of feature maps from the teacher and student models. $\mathcal{L}_R(\cdot)$ indicates the correlation function between the teacher and student feature maps.

Example (1/3)

- Junho et al. proposed a typical example of relation-based knowledge distillation.
- Introduced the concept of FSP (flow of the solution procedure) matrix.



Junho Yim et al. (2017). "A gift from knowledge distillation: Fast optimization, network minimization and transfer learning". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4133–4141

Example (2/3)

The FSP matrix $G \in R^{m \times n}$ is generated by the features from two layers. Let one of the selected layers generate the feature map $F^1 \in R^{h \times w \times m}$, where h, w , and m represent the height, width, and number of channels, respectively. The other selected layer generates the featuremap $F^2 \in R^{h \times w \times n}$. Then, the FSP matrix $G \in R^{m \times n}$ is calculated by

$$G_{i,j}(x; W) = \sum_{s=1}^h \sum_{t=1}^w \frac{F_{s,t,i}^1(x; W) \times F_{s,t,j}^2(x; W)}{h \times w}$$

where x and W represent the input image and the weights of the DNN, respectively.

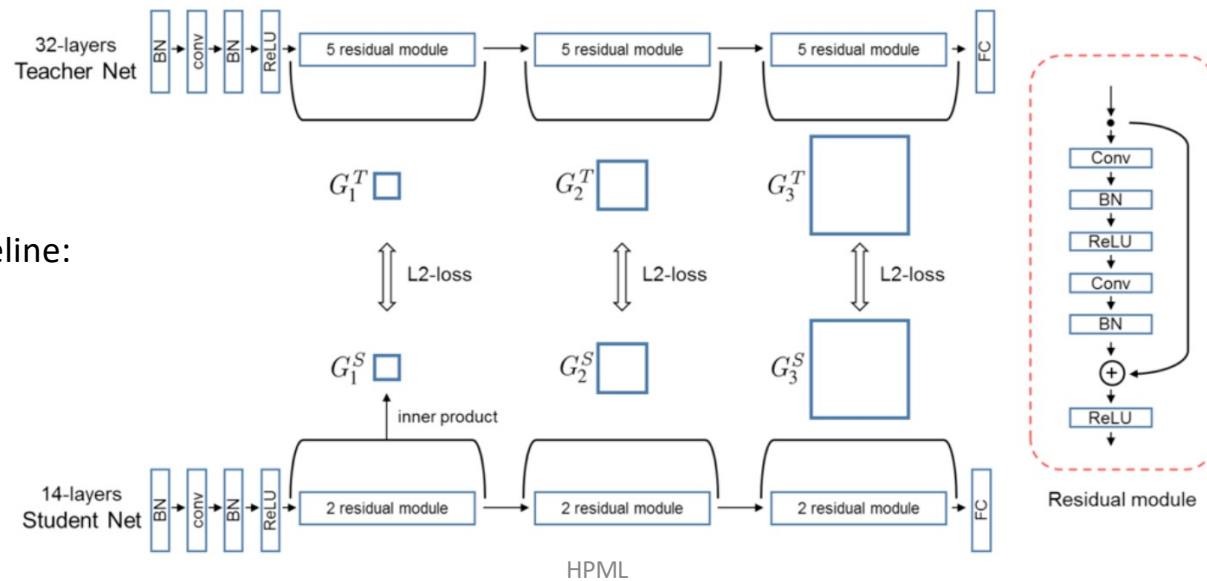
Example (3/3)

Suppose the FSP matrices of teacher network and student network are defined as $G^T(x; W_t)$ and $G^S(x; W_s)$, the knowledge distillation loss is then calculated as:

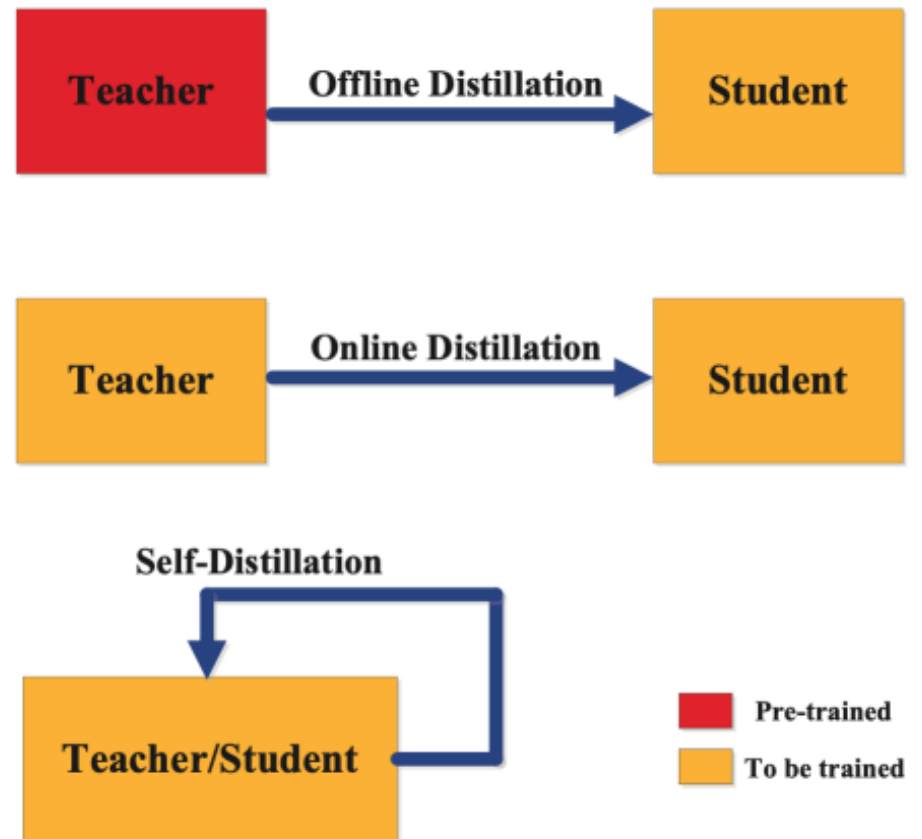
$$\mathcal{L}_{KD}(W_t, W_s) = \frac{1}{N} \sum_x \sum_{i=1}^n \lambda_i \times \|G_i^T(x; W_t) - G_i^S(x; W_s)\|_2^2$$

where λ_i and N represent the weight for each loss term and the number of data points, respectively.

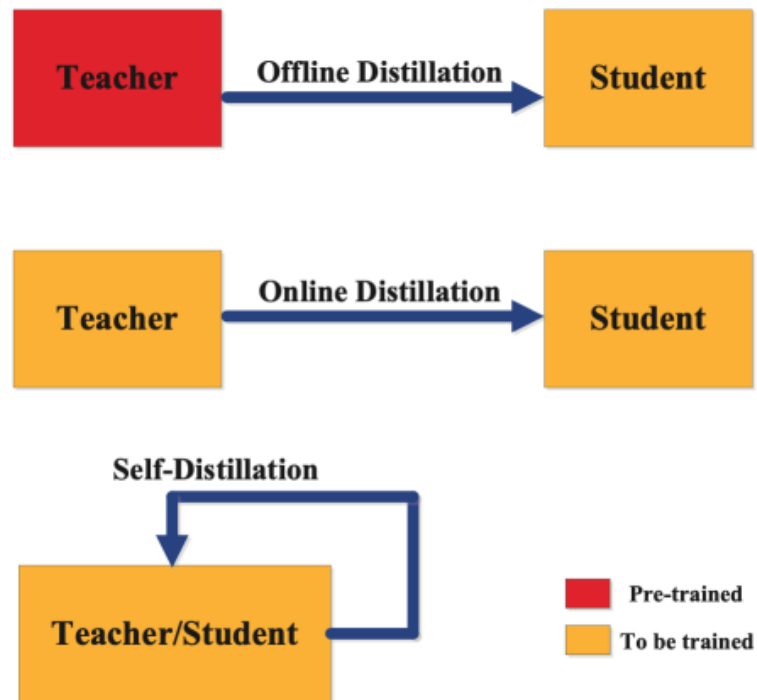
Network Pipeline:



Distillation Methods



Types of Knowledge Distillations Learning Schemes



Source: [Arxiv](#)

Offline Distillation



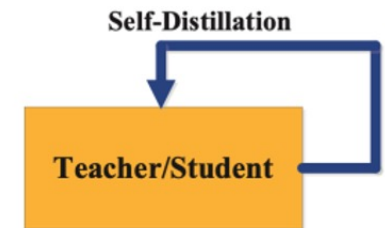
- Most common method where a pre-trained teacher model guides the student model.
- Teacher model is pre-trained on a dataset before knowledge is distilled to train the student model.
- Wide variety of pre-trained neural network models available for different use cases.
- Established technique in deep learning and relatively easier to implement.

Online Distillation



- Offline Distillation:
 - A pre-trained teacher model is typically a large-capacity deep neural network.
 - Some use cases may lack pre-trained models for offline distillation.
- Online Distillation:
 - Teacher and student models are updated simultaneously in a single end-to-end training process.
 - Operationalized using parallel computing for high efficiency.

Self-Distillation



- The same model is used for both teacher and student models.
- Knowledge from deeper layers of deep neural networks is used to train shallow layers.
- A special case of online distillation, instantiated in various ways.
- Transfer of knowledge from earlier epochs of the teacher model to later epochs for student model training.

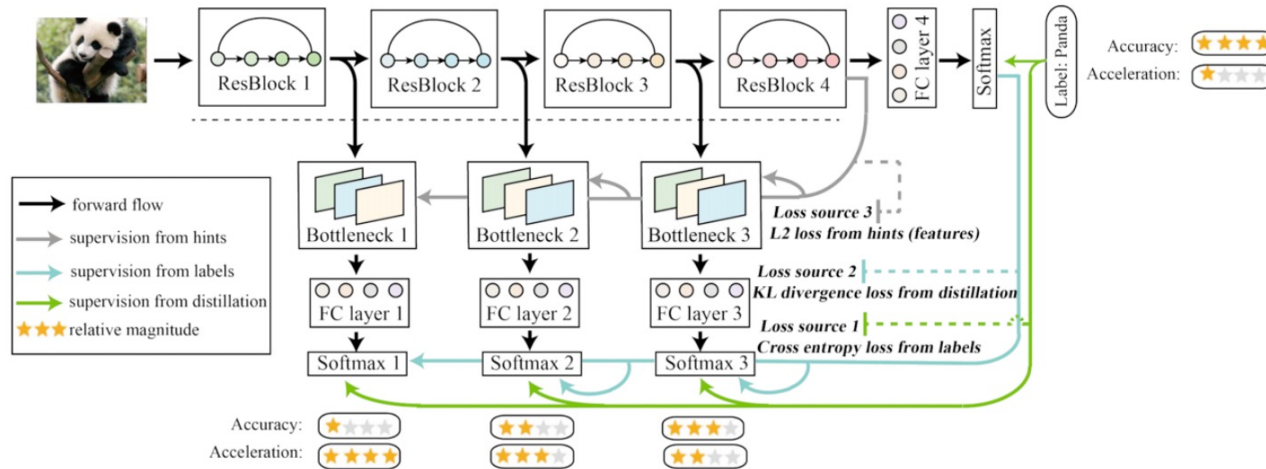
Example

- Deploying different versions of the same model on various of a mobile device
 - Early Exits
- Knowledge Distillation from the Final Exit Model can be used to improve the accuracy of the Early Exit models



Example from [Mohamed Abdelfattah](#), Cornell

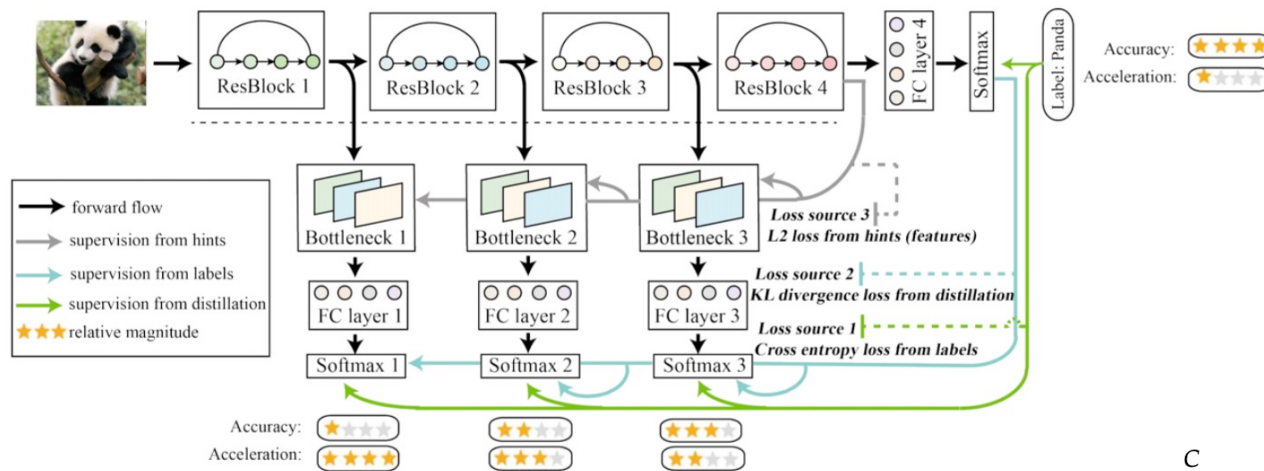
Example



Knowledge from the deeper sections of the network is distilled into its shallow sections.

Linfeng Zhang et al. (2019). “Be your own teacher: Improve the performance of convolutional neural networks via self distillation”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 3713–3722

Example



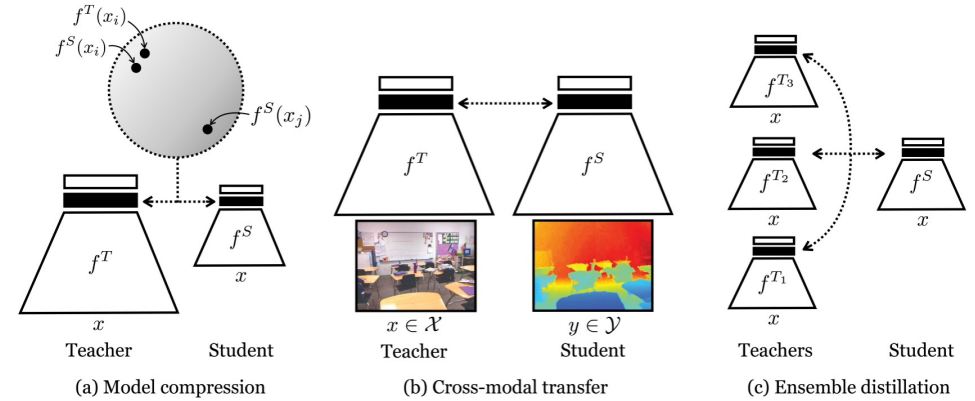
Distillation loss:

- α : Hyper-parameter
- i : i_{th} sub-network
- C : Number of sub-network
- q^i : Logits of i_{th} sub-network
- F_i : Features of i_{th} sub-network
- $\mathcal{L}_{CE}$: Cross entropy loss
- $\mathcal{L}_{KL}$: KL divergence loss
- y : Ground truth

$$\mathcal{L}_R = \sum_i^C ((1 - \alpha) \cdot \mathcal{L}_{CE}(q^i, y) + \alpha \cdot \mathcal{L}_{KL}(q^i, q^C) + \lambda \cdot \|F_i - F_C\|_2^2)$$

Linfeng Zhang et al. (2019). “Be your own teacher: Improve the performance of convolutional neural networks via self distillation”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 3713–3722

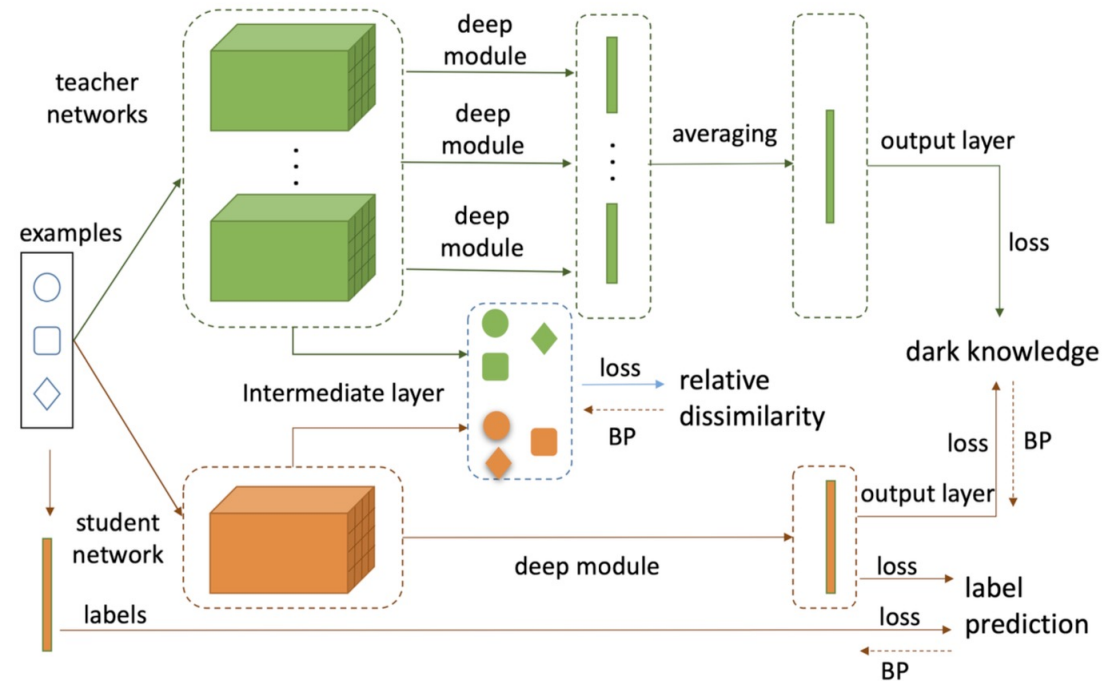
Distillation Scenarios



<https://hobbitlong.github.io/CRD/>

Ensemble KD

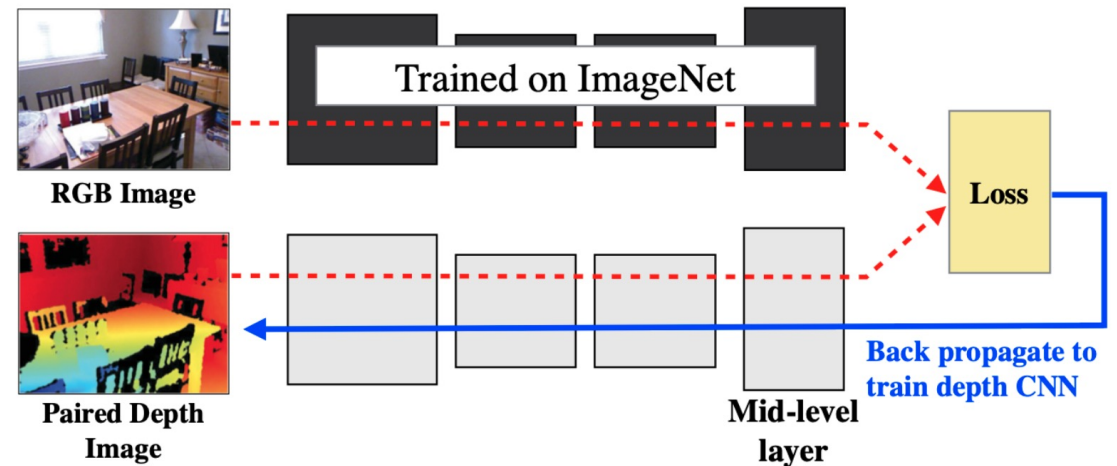
- The proposed method involves training a new, thin, deep student network by incorporating multiple comparable teacher networks.
- Three losses were utilized:
 - Label prediction loss
 - Dark knowledge loss
 - Relative similarity loss
- Incorporation of multiple teacher networks occurs in:
 - Output layers: Softened output targets are averaged.
 - Intermediate layer: Best triplet ordering relationships are determined.



Shan You et al. (2017). "Learning from multiple teacher networks". In: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1285–1294.

Cross-Modal KD

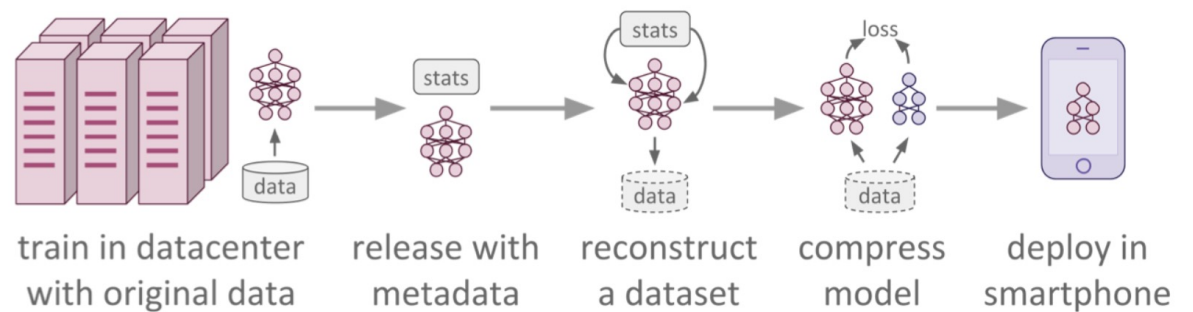
Combine KD and Cross-Modal Learning. Architecture: We train a CNN model for a new image modality (like depth images) by teaching the network to reproduce the mid-level semantic representations learned from a well-labeled image modality (such as RGB images) for modalities for which there are paired images.



Saurabh Gupta, Judy Hoffman, and Jitendra Malik (2016). “Cross modal distillation for supervision transfer”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2827–2836

Data-free KD

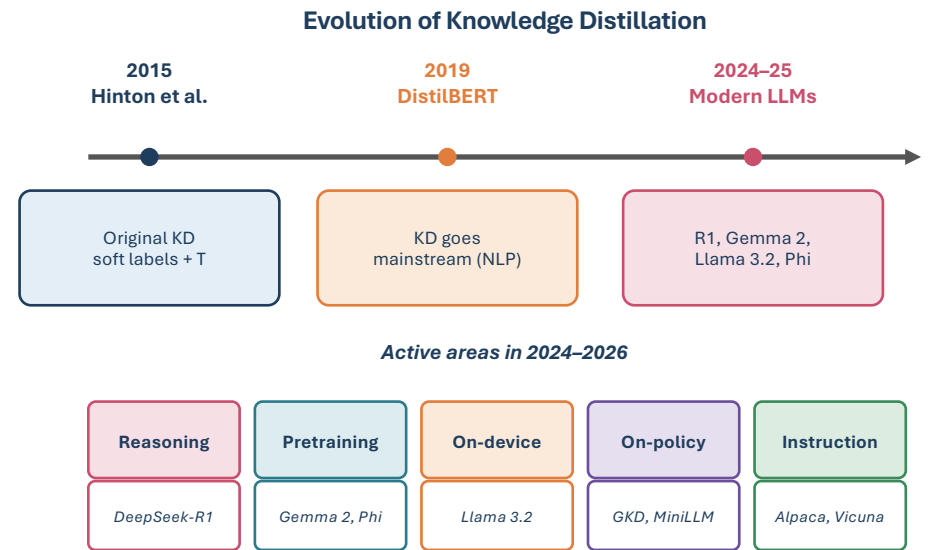
Combine KD and Data Free Compression. The proposed model compression pipeline: a model is trained in a datacenter and released with metadata. Then, another entity uses that metadata to reconstruct a dataset, which is then used to compress the model with Knowledge Distillation. Finally, the model is deployed in a smartphone.



Raphael Gontijo Lopes, Stefano Fenu, and Thad Starner (2017). “Data-free knowledge distillation for deep neural networks”. In: *arXiv preprint arXiv:1710.07535*.

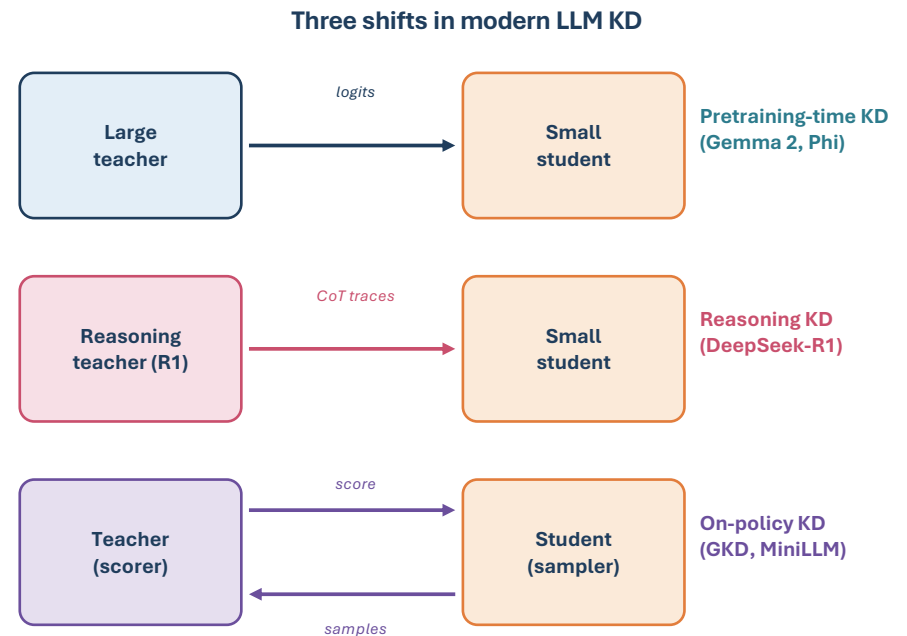
KD in Modern LLM Stacks

- Knowledge distillation is now central to nearly every open-weight LLM release.
- This section covers the techniques used in production today:
 - DeepSeek-R1 reasoning distillation (Jan 2025)
 - Llama 3.2 – pruning + distillation for on-device
 - Gemma 2 and Phi – pretraining-time distillation
 - On-policy distillation (GKD, MiniLLM)
 - Chain-of-thought / reasoning distillation
 - End-to-end pipeline in modern LLM training



The Modern LLM Distillation Landscape

- Three big shifts since 2023:
 - **Pretraining-time distillation** – small model trained from scratch with teacher's distribution as target (Gemma 2, Phi)
 - **Reasoning distillation** – capture step-by-step reasoning traces, not just answers (DeepSeek-R1)
 - **On-policy distillation** – sample from student to fix train/inference mismatch (GKD, MiniLLM)
- **Why KD is now critical for LLMs:**
 - Frontier teachers (70B–700B) are too costly to deploy at scale
 - Small students (1B–8B) hit accuracy targets at a fraction of cost
 - Pretraining data is increasingly synthetic teacher output – KD-style training is converging with vanilla pretraining



Modern LLM KD — Three Shifts in Detail

From post-hoc compression to a core training principle

1. PRETRAINING-TIME KD

Train the small model from scratch using teacher's distribution

$$L = \text{KL}(p_T(\cdot|\text{context}) \parallel p_S(\cdot|\text{context}))$$

vs. $L = -\log p_S(y_{\text{true}}|\text{context})$ [standard pretraining]

Per-token signal: ≈ 16 bits/token (full vocab distribution)

vs. 1 bit/token (one-hot ground truth)

Notable examples

Gemma 2 (9B, 27B) · Google, 2024

Phi-3 / Phi-4 · Microsoft, 2024

Result: Gemma 2 9B matches
Gemma 1 27B at $\sim 3\times$ less compute

- + Best per-token signal
- Must run teacher fwd pass over full corpus

2. REASONING KD

Transfer step-by-step reasoning via SFT on CoT traces

$$L = -\sum_t \log p_S(y_t | y_{<t}) \text{ (standard CE on traces)}$$

Targets: full reasoning trace, not just final answer

Per-token signal: Trace length: 200–2000 tokens typical

R1 used 800K curated traces

Notable examples

DeepSeek-R1 distill · Jan 2025

STaR / Self-Taught · Bootstrapping

Result: R1-Distill-Qwen-7B beats GPT-4o
on AIME, MATH (math reasoning)

- + Cheap (just SFT, no RL)
- Student may imitate without truly reasoning

3. ON-POLICY KD

Student samples; teacher scores. Fixes train/inference mismatch

$$L = D(p_T(y_s|\cdot) \parallel p_S(y_s|\cdot)), y_s \sim p_S$$

$D \in \{\text{forward KL, reverse KL, JS, generalized JS}\}$

Per-token signal: Reverse KL: mode-seeking $\rightarrow$ cleaner outputs

Forward KL: mean-seeking $\rightarrow$ averages modes

Notable examples

GKD (Agarwal 2023) · Generalized KD

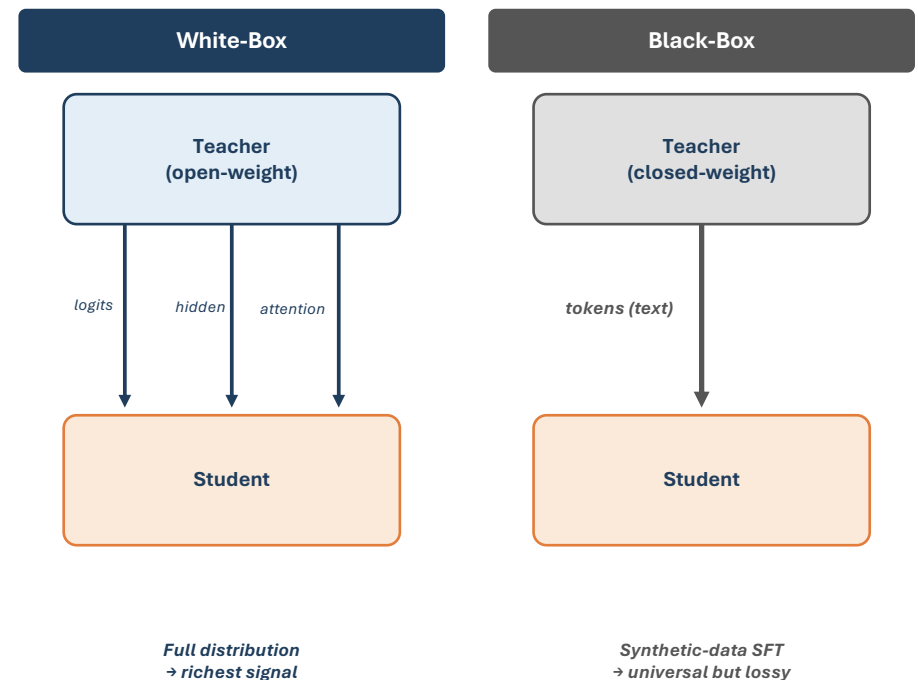
MiniLLM (Gu 2023) · Reverse-KL focused

Result: Used in Gemma instruction tuning,
Llama 3 fine-tuning

- + No exposure bias
- 1.5–2x compute per training step

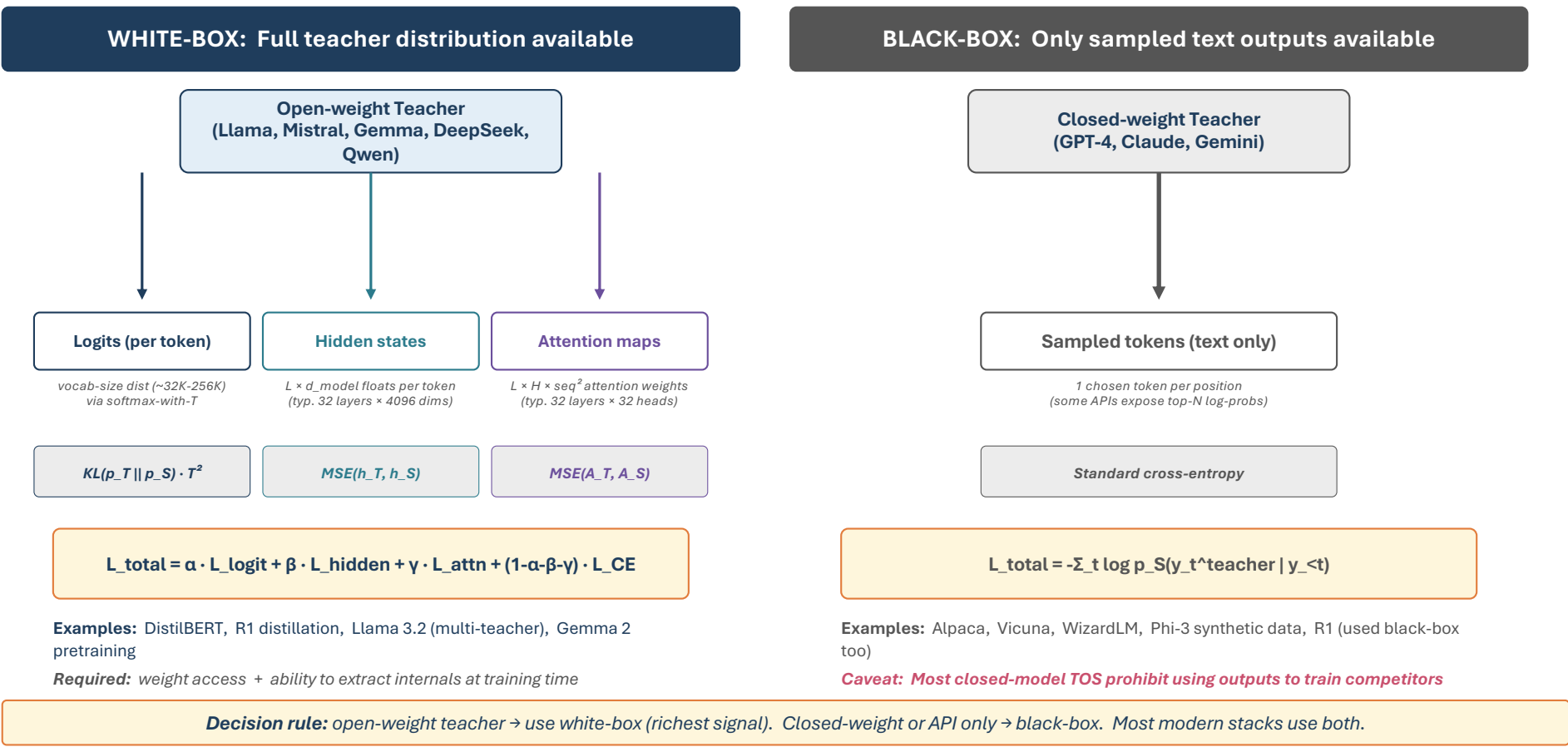
White-Box vs Black-Box Distillation

- **White-box:** student has access to teacher's full output distribution
 - KL divergence on full vocab logits, hidden state matching, attention transfer
 - Requires open-weight teacher (Llama, Mistral, Gemma, DeepSeek, Qwen)
 - Richest signal — full distribution carries dark knowledge per token
- **Black-box:** student only sees teacher's text outputs (sampled tokens)
 - Equivalent to training on synthetic data with cross-entropy
 - Used for closed-weight teachers (GPT-4, Claude, Gemini)
 - Examples: Alpaca, Vicuna, WizardLM, Phi-3 synthetic data
- Modern stacks combine both — but check teacher TOS for closed-weight models



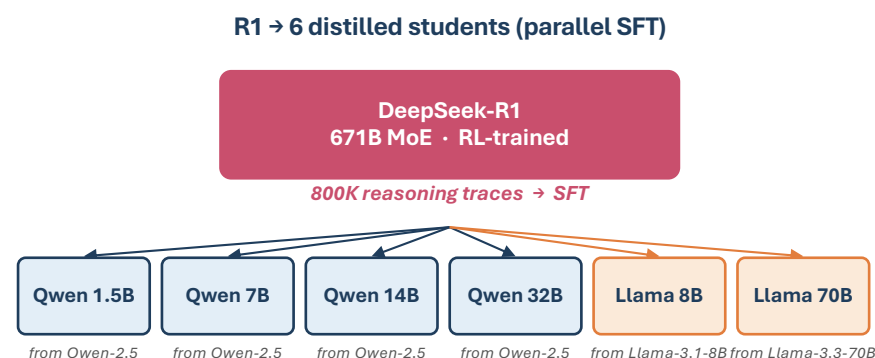
White-Box vs Black-Box KD — Decision Guide

What can you read from the teacher?



DeepSeek-R1: Reasoning Distillation (Jan 2025)

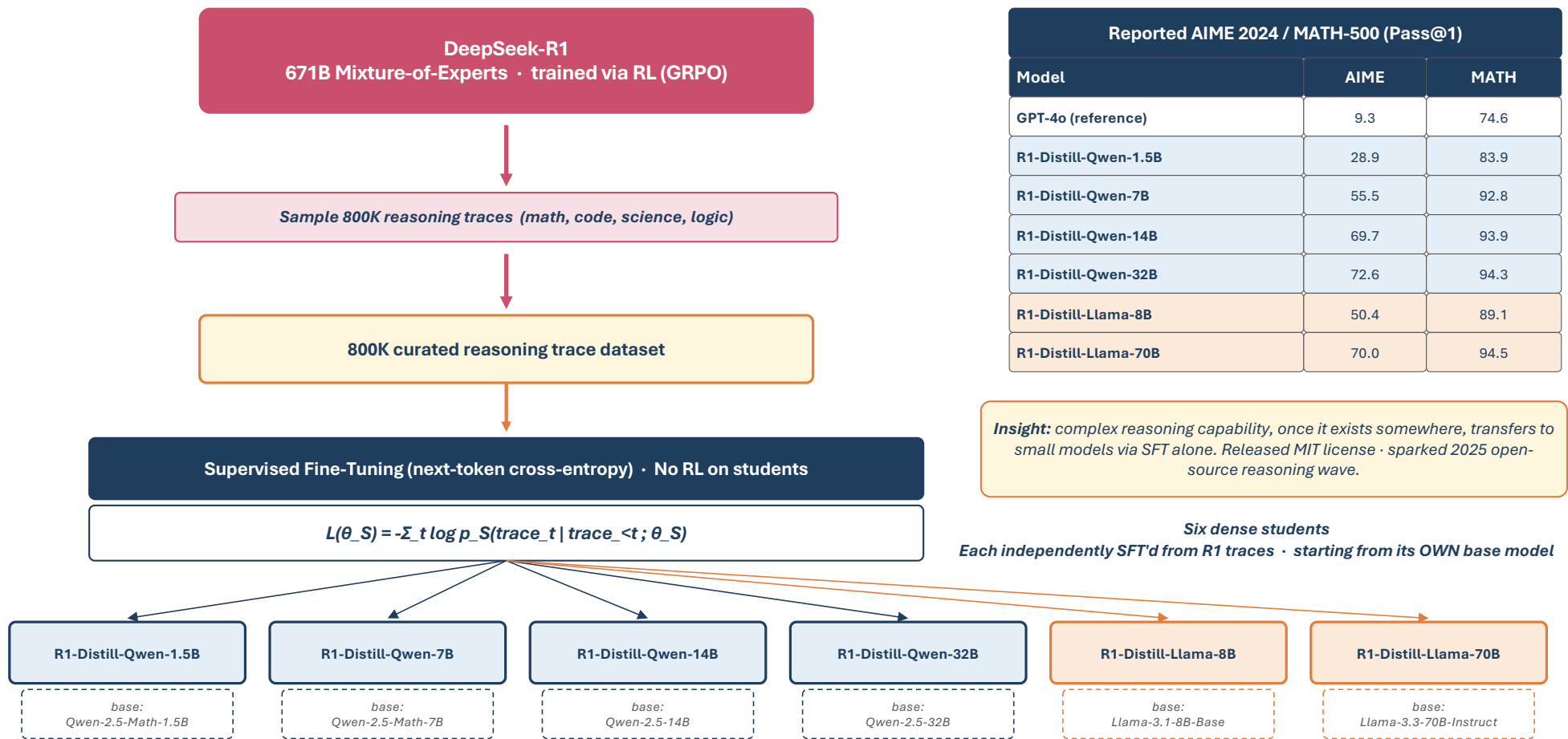
- **DeepSeek-R1** — 671B Mixture-of-Experts reasoning model trained largely via reinforcement learning
- **Distilled into 6 dense students:** Qwen-1.5B, 7B, 14B, 32B, Llama-8B, 70B
- **Method:** pure supervised fine-tuning of students on 800K reasoning traces from R1
 - No RL on the students — black-box CoT distillation alone was sufficient
- **Results:** R1-Distill-Qwen-7B beats GPT-4o on math (AIME, MATH); 32B variant approaches o1-mini
- **Key insight:** complex reasoning capabilities CAN be transferred via supervised distillation alone
- Released under MIT license — sparked the 2025 open-source reasoning model wave



*All 6 students distilled directly from R1 — not chained.
No RL on students; pure supervised CoT distillation.*

DeepSeek-R1 Distillation — Full Picture

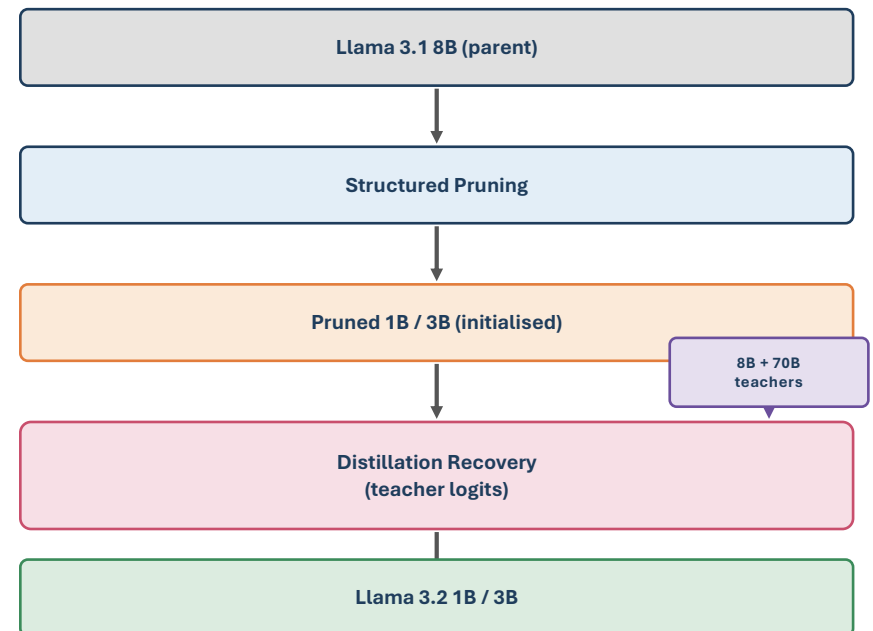
All 6 students fine-tuned on top of pre-existing base models, in parallel, from the same 800K R1 traces



Llama 3.2 (1B / 3B): Pruning + Distillation

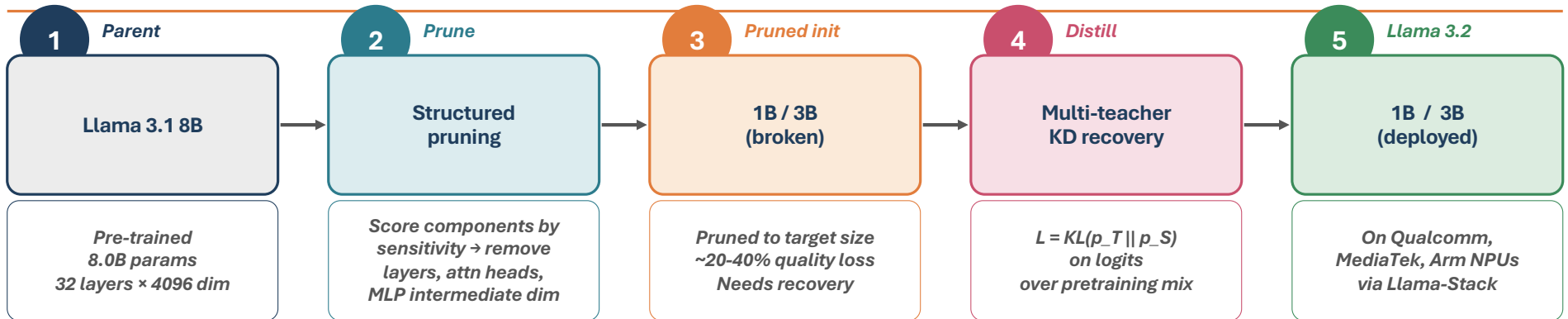
- Targets on-device deployment — phones, embedded systems, edge inference
- **Pipeline:**
 - Start from Llama 3.1 8B as the parent
 - Apply structured pruning to remove layers and reduce hidden dim to hit 1B / 3B targets
 - Use Llama 3.1 8B AND 70B as teachers — distillation losses based on logits during recovery
 - Continue pretraining on standard data with distillation loss as the training objective
- Recovers most of the parent's quality at a fraction of the size
- Deployed via Llama-Stack on Qualcomm, MediaTek, and Arm-based mobile platforms
- Now a standard recipe — NVIDIA's Minitron paper independently arrived at the same approach

Llama 3.2 — prune-then-distill pipeline

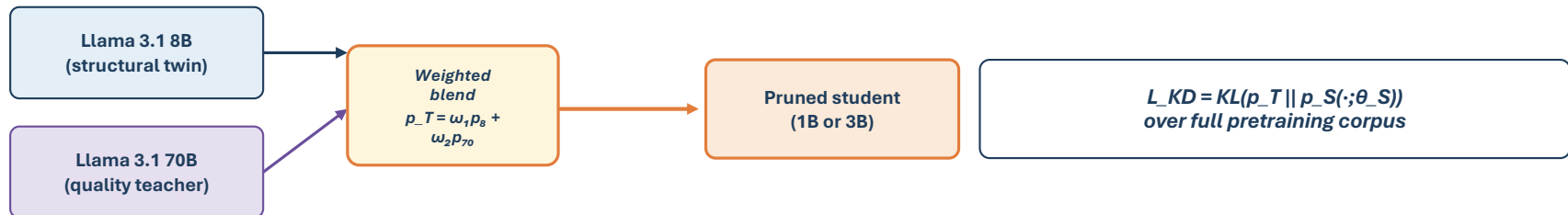


Llama 3.2 Pipeline — Prune-then-Distill

Industry standard recipe (Meta, NVIDIA Minitron) for on-device LLMs



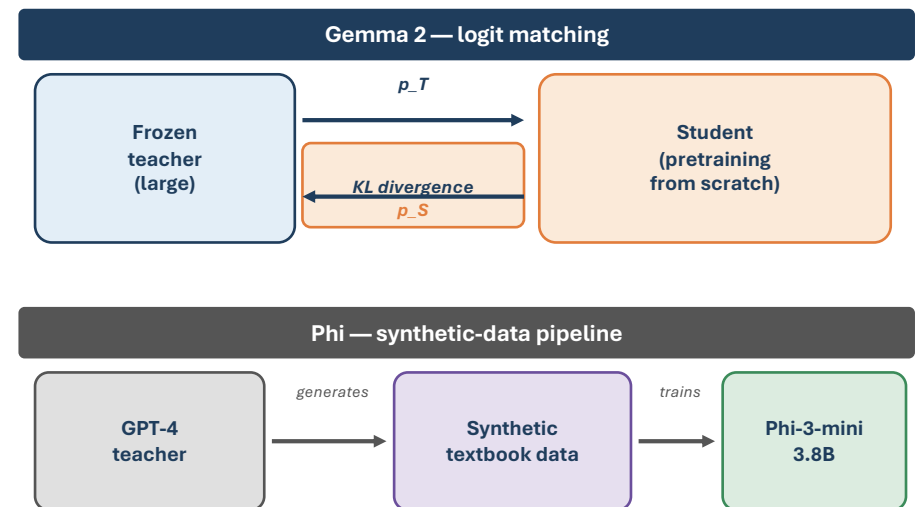
Detail: stage 4 — multi-teacher distillation recovery



Why prune-then-distill beats both alternatives: starting from pruned 8B inherits useful structure (vs random init); pruning gives architectural flexibility (vs fixed-architecture distillation alone). NVIDIA's Minitron paper independently arrived at the same recipe.

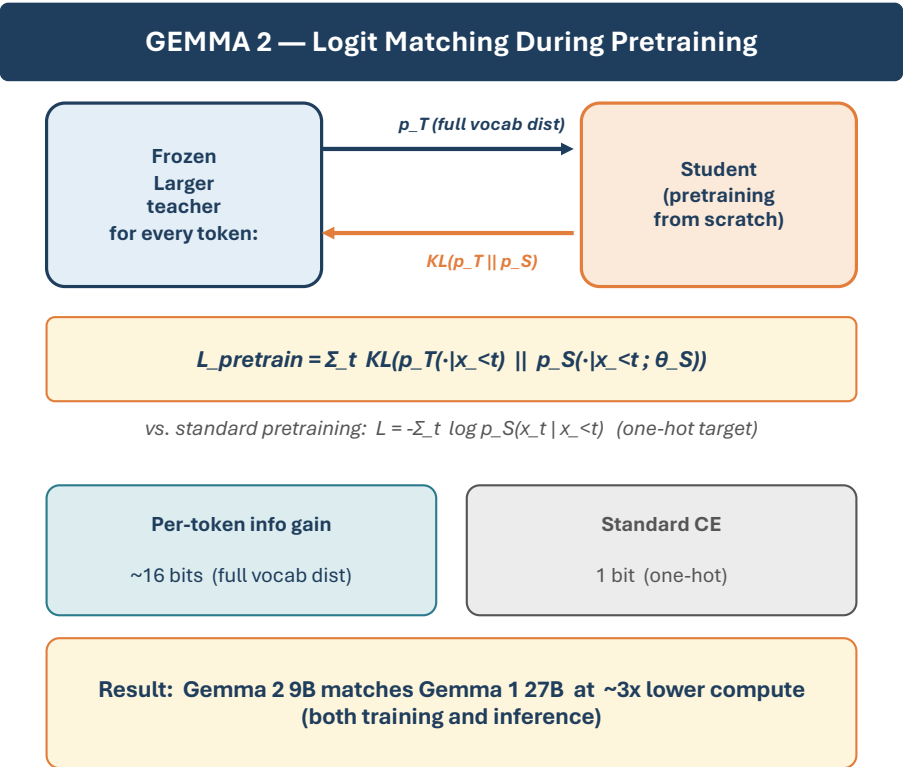
Gemma 2 & Phi: Pretraining-Time Distillation

- **Gemma 2 (Google, 2024)**
 - 9B and 27B models pretrained with KD from a much larger teacher
 - Standard next-token cross-entropy replaced by KL divergence vs. teacher's full distribution
 - Result: Gemma 2 9B matches prior generation's 27B at ~3x lower compute
- **Phi-3 / Phi-4 (Microsoft) – "Textbooks Are All You Need"**
 - Strong teacher (GPT-4) generates curated synthetic data – textbook-style explanations, code, reasoning
 - Small student trained on this synthetic corpus with standard pretraining loss
 - Phi-3-mini (3.8B) competitive with Mixtral-8x7B on many benchmarks
- Bottom line: distillation has moved up the stack – from post-hoc compression to a core training method

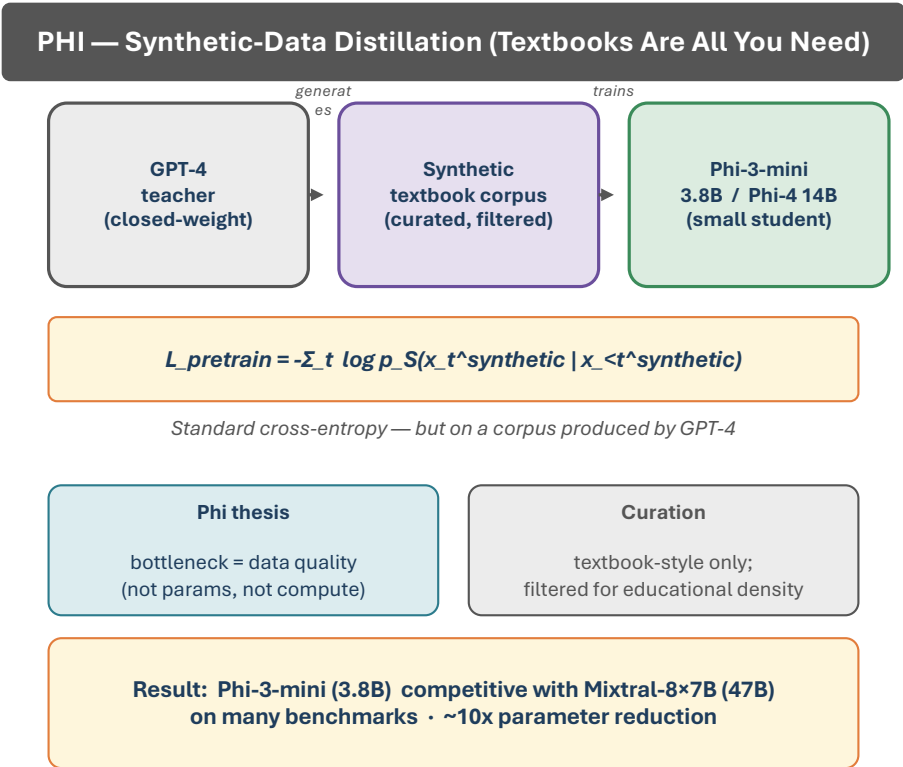


Pretraining-Time KD — Two Recipes Compared

Both shift KD upstream into pretraining; very different mechanisms



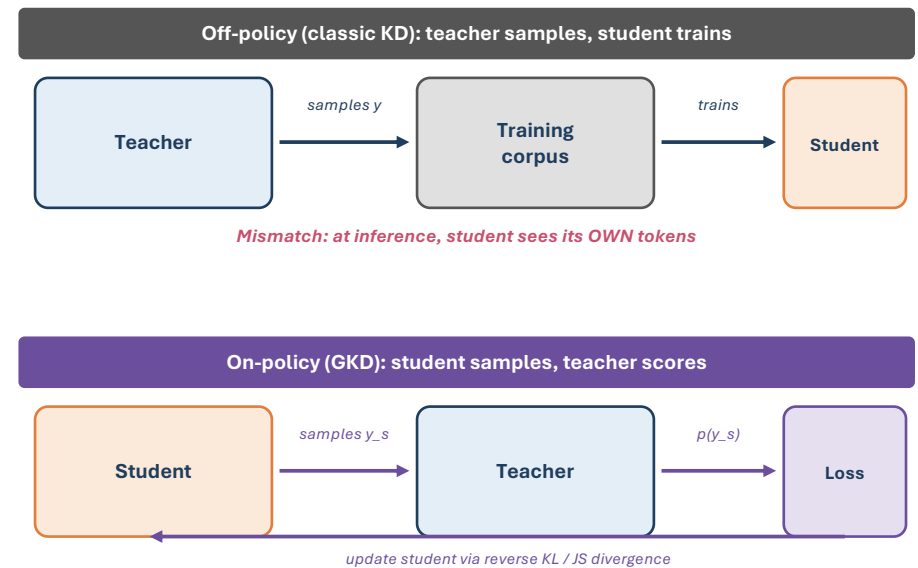
Type: WHITE-BOX · requires teacher-fwd-pass cost over full corpus



Type: BLACK-BOX · works with any teacher accessible by API

On-Policy Distillation (GKD, MiniLLM)

- **Problem: train-inference distribution mismatch**
 - Classic KD: student trained on teacher's tokens; at inference, sees its own tokens
 - This is exposure bias for distillation — hurts at scale
- **Generalized Knowledge Distillation (GKD, Agarwal et al., 2023)**
 - Sample sequences ON-POLICY from the student
 - Score with teacher distribution at each student-sampled token
 - Match using divergence: forward KL, reverse KL, JS, or generalized JS
- **Reverse KL is mode-seeking** — student picks one good answer instead of averaging across modes
- Used in Gemma instruction tuning, MiniLLM, Llama 3 fine-tuning, and most modern instruction-distilled LLMs



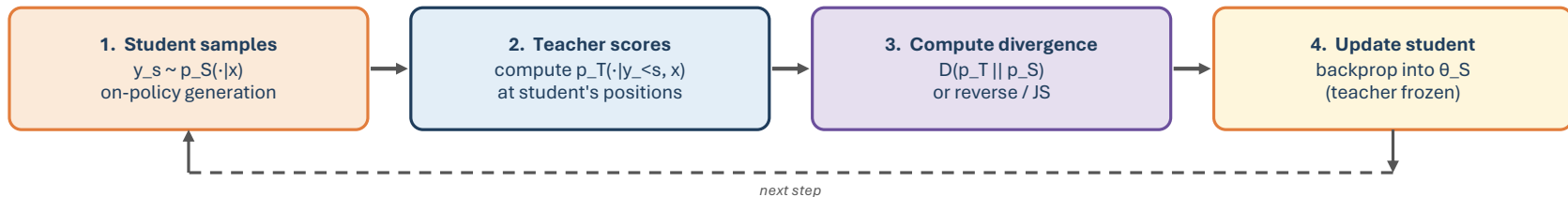
On-Policy Distillation (GKD) — Why It Matters

Fixing the train/inference distribution mismatch in classical KD

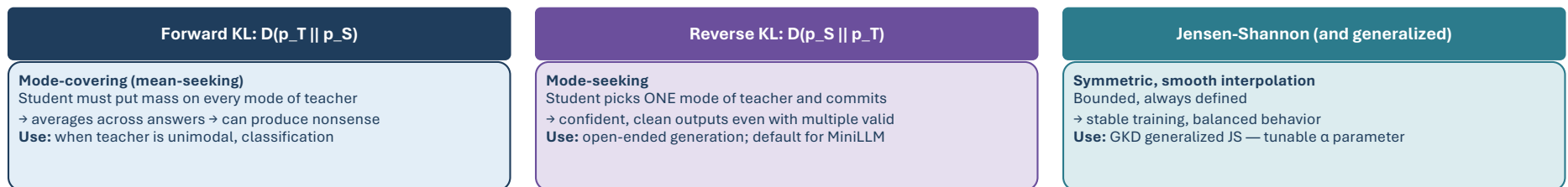
The problem: exposure bias



The fix: sample from student, score with teacher



Divergence choice: fundamentally changes student behavior



Chain-of-Thought / Reasoning Distillation

- Goal: transfer reasoning ability — not just final answers
- Method: teacher generates step-by-step traces; student SFT on those traces as targets
- Variants worth knowing:
 - **Distilling Step-by-Step** (Hsieh et al., 2023) — rationales as auxiliary loss; small student matches larger one with less data
 - **Symbolic CoT** — extract structured reasoning steps (variables, equations) instead of free-form text
 - **Self-distilled CoT** — model bootstraps better reasoning from its own outputs (STaR-style)
- DeepSeek-R1 distillation is the most prominent CoT distillation success at scale
- **Open question:** do small students truly LEARN to reason, or just imitate surface patterns? Recent evidence is encouraging.

Reasoning trace as the supervision signal

Teacher trace

Q: $23 \times 17 = ?$
Step 1: $23 \times 10 = 230$
Step 2: $23 \times 7 = 161$
Step 3: $230 + 161 = 391$
A: 391

SFT on traces

Student (after distillation)

Q: $47 \times 13 = ?$ (new problem)
Step 1: $47 \times 10 = 470$
Step 2: $47 \times 3 = 141$
Step 3: $470 + 141 = 611$
A: 611

Student learns reasoning STRUCTURE, not just answers

Chain-of-Thought Distillation — Detailed Walk-through

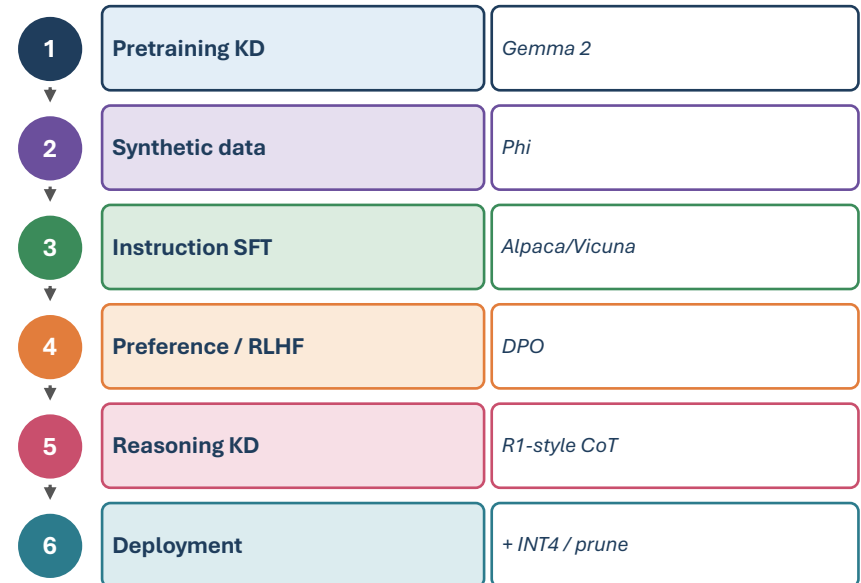
Why teaching the student the *PROCEDURE* generalizes; teaching only *ANSWERS* doesn't

ANSWER-ONLY KD (fails)	COT DISTILLATION (R1-style)	DISTILLING STEP-BY-STEP
<i>Training pair:</i> Q: $23 \times 17 = ?$ A: 391	<i>Training pair:</i> Q: $23 \times 17 = ?$ Step 1: $23 \times 10 = 230$ Step 2: $23 \times 7 = 161$ Step 3: $230 + 161 = 391$ A: 391	<i>Multi-task training:</i> Q: $23 \times 17 = ?$ [label head]: 391 [rationale head]: Step 1: $23 \times 10 = 230$ Step 2: $23 \times 7 = 161$ Step 3: $230 + 161 = 391$
<i>On novel problem:</i> Q: $47 \times 13 = ?$ Student prediction: 611 ✗ (memorized — no procedure) <i>Student learns $Q \rightarrow A$ mapping. No internal procedure. Fails on out-of-distribution problems.</i>	<i>On novel problem:</i> Q: $47 \times 13 = ?$ Step 1: $47 \times 10 = 470$ Step 2: $47 \times 3 = 141$ Step 3: $470 + 141 = 611$ A: 611 ✓ <i>Student learns the <i>PROCEDURE</i> for multiplication, encoded as structured-trace generation.</i>	<i>Effective sample efficiency:</i> Small student matches 10x larger teacher with ~10x less data (Hsieh et al., 2023) <i>Two heads: predict label AND rationale. Both regularize each other → strong sample efficiency.</i>
✗ Doesn't generalize	✓ Generalizes via procedure	✓ Both generalizes & data-efficient

KD Pipeline in a Modern LLM Stack

- **1. Pretraining** — distill smaller foundation model from larger teacher (Gemma 2 style)
- **2. Synthetic data generation** — teacher generates high-quality SFT data (Phi style)
- **3. Instruction distillation** — SFT student on teacher-generated instruction-following data (Alpaca / Vicuna)
- **4. Preference / RLHF distillation** — distill teacher's reward signal or policy (DPO and variants)
- **5. Reasoning distillation** — SFT on teacher CoT traces (R1 style)
- **6. Deployment** — serve compact student, often combined with INT4/INT8 quantization and pruning
- **Practical advice:** start with response-based KD; add feature-based or on-policy only if it plateaus. Most production wins come from good data, not exotic techniques.

Six-stage modern LLM distillation pipeline



Modern LLM Distillation Pipeline — End to End

Six stages where KD appears in the modern training stack — each with technique, concrete examples, and tradeoffs

1	Pretraining KD Train student from scratch with teacher's full distribution as target	white-box	Examples: Gemma 2 (9B, 27B)
	$L = KL(p_T(\cdot context) p_S(\cdot context))$	Compute: ★★★★★	Quality gain: ★★★★★
2	Synthetic data generation Generate textbook-quality corpus from a strong teacher	black-box	Examples: Phi-3 / Phi-4, OpenLLaMA
	$L = \text{standard cross-entropy on synthetic corpus}$	Compute: ★★★★	Quality gain: ★★★★
3	Instruction distillation SFT on teacher-generated instruction-following pairs	black-box	Examples: Alpaca, Vicuna, WizardLM
	$L = -\sum \log p_S(\text{response}_t \text{response}_{<t}, \text{instruction})$	Compute: ★★	Quality gain: ★★★
4	Preference / RLHF distillation Distill teacher's reward signal or aligned policy (DPO, IPO)	white or black	Examples: Llama 3 Instruct, Zephyr
	$L_{DPO} = -\log \sigma(\beta \log p_S(y_w) / p_{ref}(y_w) - \beta \log p_S(y_l) / p_{ref}(y_l))$	Compute: ★★★	Quality gain: ★★★★★
5	Reasoning distillation SFT on chain-of-thought traces from a reasoning teacher	black-box	Examples: DeepSeek-R1 distill, STaR
	$L = -\sum \log p_S(\text{trace}_t \text{trace}_{<t})$	Compute: ★★	Quality gain: ★★★★★ (math/code)
6	Deployment compression Combine with quantization (INT4/INT8) and pruning for hardware fit	structural	Examples: Llama 3.2 + INT4, Phi-3-mini-4k-instruct-Q4
	$L_{QAT} + \text{pruning regularizer}$	Compute: ★	Quality gain: ★★★

Example Notebook

- PyTorch KD Tutorial:
https://pytorch.org/tutorials/beginner/knowledge_distillation_tutorial.html

Further Readings

- Hinton et al., Distilling the Knowledge in a Neural Network, <https://arxiv.org/abs/1503.02531>
- Jianping Gou et al., Knowledge Distillation: A Survey, <https://arxiv.org/abs/2006.05525>
- DeepSeek-AI, **DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning**, <https://arxiv.org/abs/2501.12948>
- Gemma Team, **Gemma 2: Improving Open Language Models at a Practical Size**, <https://arxiv.org/abs/2408.00118>
- Agarwal et al., **On-Policy Distillation of Language Models (GKD)**, <https://arxiv.org/abs/2306.13649>
- Hsieh et al., **Distilling Step-by-Step!**, <https://arxiv.org/abs/2305.02301>